QUALIFICATION STANDARD AND PLAN (QSP)

Weapons Engineering Technician (W ENG Tech)
NQual (AKQD)

MITE ID
Residential Phase – 118940

Regular Force: Weaspons Engineering Technician 00366 (W ENG Tech)
Training Authority (TA): Commander Naval Training Group (CNTG)
Curriculum Control Authority (CCA): NFS(P)

MARPAC RDIMS #1424172 v2
Chap 1 & 2 approved 09/09/ 2024
Chap 3 & 4 approved 09/09/ 2024

RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.0		21 Feb 11	DMTE 2-4-2
2.0		Jun 14	Cdr D.J. Benoit CFNES Cmdt
Version 2	- QS amendment -Change to new format Addition of Unit Technical Inspector MITE 302486	Sept 2024	NTG HQ
Vesion 2	-Tp amendment	Sept 2024	CCA.

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the NTG Website. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Training Group (CNTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon following:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-002 Non-Commission Member General Specification dated 09 February 2011; and

   c. A-PD-055-002/PP-001 Weapons Engineering Technician 00366 dated 01 September 2011.

3. This QSP publication is effective on receipt and supersedes AKQD QSP revision 2.0, dated June 2014. This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened 15 Jan 2023 working in collaboration with subject matter experts drawn from participating organizations. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened Apr 2023 at NFS(P) working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a QSP Deviation Request is required.

QUALIFICATION STANDARD

MARPAC RDIMS #1424172 v2
Chap 1 & 2 approved DD MM YYYY
Chap 3 & 4 approved DD MM YYYY

CHAPTER 1 QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for the W Eng Tech to perform their duties as a W Eng Tech Armament Maintainer. Possession of this qualification will allow personnel to operate, troubleshoot, and perform preventative and corrective maintenance on ship's armaments.

GRANTING OF NQUAL

2. A trainee will achieve the qualification AKQD (MITE 118940) and W Eng Tech Armament Maintainer when the standards listed in the Performance Objectives (POs) have been met.

CO-QUALIFICATION

3. In addition to the performance objectives specific to this training, the following additional qualifications are included in this training program to meet the requirements of the job. The MITE and NQuals associated with these qualifications shall be granted upon completion of the training program and shall be listed within the course report and will be delivered IAW the following:

a. DRMIS, MITE 119097; and

b. Specialist Technical Inspector, MITE 302486.

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

a. AKNV_ Jr W Eng Tech EQ2 MITE 118939 or MITE 123762; and

b. AKPK_W Eng RQS1 Academics MITE 118966.

CHAPTER 2 PERFORMANCE OBJECTIVES

PO 001 Maintain the Multi Ammunittion Softkill System

1. Performance: Maintain the MASS.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) MASS and/or MPT

      (6) PPE; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental.

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The W Eng Armament System Maintainer shall maintain the MASS, IAW refs, by;

   a. describing the MASS;

   b. initializing the MASS;

   c. operating the MASS; and

   d. performing pre/post firing check on the MASS.

4. References:

   a. A1 – A7; and

b. E1 – E10.

5. Task Numbers:

a. E001 – E004.

6. Remarks: Nil.

PO 002 Maintain the Ship's Main Gun System

1. Performance: Maintain ship's main gun system.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Main Gun System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The W Eng Armament System Maintainer shall maintain the ship's main gun system, IAW refs, by:

   a. operating the ship's main gun system, to include:

      (1) performing close-up clear away

      (2) powering up; and

      (3) carrying out system functional check(s).

   b. performing PM on the ship's main gun system, to include:

      (1) interpreting CFTO;

      (2) performing PM IAW CFTO; and

(3) completing maintenance administration.

c. performing CM on the ship's main gun system, to include:

(1) identifying deficiency(s);

(2) performing diagnostics;

(3) conducting fault finding;

(4) identifying failed LRU;

(5) replacing LRU;

(6) performing functional check(s); and

(7) completing maintenance administration.

4. References:

a. A1 – A7; and

b. B1 – B12.

5. Task Numbers:

a. C001 - C018.

6. Remarks: Nil.

PO 003 Maintain the Close In Weapons System

1. Performance. Maintain Close In Weapons System (CIWS).

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS; and

      (7) assistance.

   b. Denied:

      (1) Nil

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night

3. Standard. The W Eng Armament System Maintainer shall maintain the CIWS, IAW refs, by:

   a. operating the CIWS, to include:

      (1) powering up;

      (2) carrying out system functional check(s); and

      (3) verifying local and remote operations.

   b. performing PM on the CIWS, to include:

      (1) interpreting CFTO;

      (2) performing PM IAW CFTO; and

      (3) completing maintenance administration.

c. performing CM on the CIWS, to include:

(1) identifying deficiency(s);

(2) performing diagnostics;

(3) conducting fault finding;

(4) identifying failed LRU;

(5) replacing LRU;

(6) performing functional check(s); and

(7) completing maintenance administration.

4. References:

a. A1 – A7; and

b. C1 – C20.

5. Task Numbers:

a. C019 - C033.

6. Remarks : Nil.

PO 004 Operate Close In Weapons System In All Modes

1. Performance. Operate Close In Weapons System (CIWS) in all modes.

2. Conditions.

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS; and

      (7) assistance.

   b. Denied:

      (1) Nil

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night

3. Standard. The W Eng Armament System Maintainer shall operate the CIWS in all modes, IAW refs, by:

   a. operating the CIWS, to include:

   b. close-up to standing to state;

   c. operate in all modes as directed; and

   d. returning to secure state.

4. References.

   a. A1 – A7; and

   b. C1 – C20.

5. Task Numbers.

   a. C034 - C039.

6. Remarks: Nil.

PO 005 Maintain Combat Management System

1. Performance. Maintain Combat Management System(CMS)

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) tools;

      (4) DWAN;

      (5) PPE;

      (6) CMS; and

      (7) assistance

   b. Denied:

      (1) Nil

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The W Eng Tech Sonar Systems Maintainer shall maintain CMS, IAW refs by:

   a. operating the CMS, to include:

      (1) powering up;

      (2) carrying out system functional checks; and

      (3) verifying remote operation.

   b. performing PM on the CMS, to include:

      (1) interpreting CFTO;

(2) performing PM IAW CFTO; and

(3) completing maintenance administration.

c. performing CM on the CMS, to include:

(1) identifying deficiency;

(2) performing diagnostics;

(3) conducting fault finding;

(4) identifying failed LRU;

(5) replacing LRU;

(6) performing functional checks; and

(7) completing maintenance administration.

4. References:

a. A1 – A7; and

b. D1 – D4.

5. Task Numbers:

a. O001 – O009

6. Remarks: Nil

PO 006 Utilize DRMIS Portal as a Maintenance Supervisor (Mod 2)

1. Performance. PO 006 shall be carried out IAW PO 002 – Utilize DRMIS Portal as a Maintenance Supervisor of the DRMIS QSP 119098.

2. Conditions:

   a. Given:

      (1) References;

      (2) Supervision;

      (3) Computer with LAN access;

      (4) Printer;

      (5) DRMIS account; and

      (6) Access to DRMIS Portal.

   b. Denied:

      (1) Nil

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. PO 006 shall be carried out IAW PO 002 of the DRMIS QSP 119098.

4. References:

   a. IAW PO 002 of the DRMIS QSP 119098.

5. Task Numbers:

   a. N/A.

6. Remarks: Nil.

TRAINING PLAN

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CHAPTER 3 TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. Enabling Checks (ECs) are formative tests that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructor(s) and trainees. Each EO must have at least one EC. Marks for ECs are not calculated in the trainee's final mark;

   b. Performance Checks (PCs) are summative tests are used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Summative tests can take the form of PCs or Critical ECs (reserved for safety related content). Supplemental PCs are administered upon failure of a PC or critical EC and must be in the same format and have the same standard as the original PC; and

   c. Use of Applicable Training Simulator software.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 106 training days.

Course Duration
Actual Training Time (ATT): 640.4 periods
Total Training Days: 106 days

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training.

Course Director				
Rank	MOSID	Qualifications Required	Qty	PO / EO

Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO

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MS/PO2	W Eng Tech Armament Instructor	Eng Tech Armement Applications, BIT	2	
PO1	W Eng Tech Manager	W Eng Tech Maintenance Manager	1	

Guest Lecturers/Others (May be civilian)				
Lecturer	# Instructor	Subject	EO #	# period

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitations are identified below:

a. Limitation 1; and

b. Limitation 2.

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

a. Optimum – 6;

b. Maximum Load – 8; and

c. Minimum – 1.

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2).

7. The authority to load members onto this course will be IAW NPTORD 4540-43. NTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by:

a. The Career Manager for career courses; or

b. NFS Campus Support for occupation and specialty courses.

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500-22. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

10. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

11. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

12. There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

13. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

14. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

15. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

16. Formal Course: A Canadian Forces Course Report, Form CF 377 and/or CF289 Certificate of Military Achievement, shall be prepared and distributed in accordance with DAOD 5031-9. The Course Report shall be descriptive, identifying the student's strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

CHAPTER 4 LESSON SPECIFICATIONS

PO 001 – Maintain the Multi Ammunition Softkill System

EO 001.01 Describe the Multi Ammunition Softkill System

1. Performance. Describe the Multi-Ammunition Softkill System (MASS).
2. Conditions:
    a. Given:
        (1) supervision;
        (2) references;
        (3) DWAN;
        (4) tools and test equipment;
        (5) MASS and/or MPT
        (6) PPE; and
        (7) assistance.
    b. Denied:
        (1) Nil
    c. Environmental:
        (1) any shipboard conditions at sea or alongside; and
        (2) day or night
3. Standard. The Trainee shall describe the MASS, IAW refs, to include:
    a. capabilities, limitations, policies and safety regulations;
    b. system reaction to threats;
    c. system functions;
    d. physical layout of equipment; and
    e. publication layout.
4. Time Required:
    a. EO –10 x 50 min periods.
5. Instructional Methods:

a. Lecture; and

b. Demonstration.

6. References:

a. A1 – A7; and

b. E1 – E10.

7. Training Aids:

a. MASS System and/or MPT;

b. audio visual aids;

c. references; and

d. handouts.

8. Test Details:

a. EO 001.01 will be tested as part of PC 001 IAW Annex K;

9. Remarks:

a. Projects and exercises may be assigned; and

b. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference

TP1	1. Describe the MASS System, to include: a. Capabilities, Limitations, policies and Safety Regulations; b. Keys; c. HERO policies; d. RADHAZ; and e. Loading. 2. Ammunition, to include: a. Types and functions; b. OMNI TRAP ER (type 1); and	L	1.5	
	c. Loading/unloading/reloading procedures. 3. Command Management System (CMS), to include; 4. Block signal flow - CMS to MASS;			
TP2	1. System reactions to threats, to include: a. Littoral threat; b. Infrared threat; and c. RF threat. d. Modes of operation: e. Manual mode; f. Integrated mode; and g. Simulation mode.	L	1	

TP3	System Functions, to include; 1. Multi-functional Display: a. Hard/ Soft keys; b. Functional Pages; and c. MASS Interface Unit 2. Control Unit, to include: a. Fire Blocking Unit; b. Hour meters; c. LC Display; d. CAN Repeater; e. Transformer; f. Servo Controller; g. Power Supply; h. PLC; i. Inclinometer; j. Fuzes (CBs); 3. Manual Panels;	L	4.5	

	a. Locations; b. Functions; 4. Launchers: a. Locations; b. Functions; c. Description; d. Base plate; e. Azimuth unit (Az); f. Elevation unit (El); g. Platform; and h. Magazine shoe and carrier. 5. Ignition system			

	1. Uninterrupted Power Supply: a. Location; b. Description; c. Capabilities; d. Power Distribution; e. Ship's power; f. System power; g. Casualty Mode (UPS Bypass); h. Maintenance Tools; i. Descriptions; j. Functions;			
TP4	1. Physical layout of equipment: a. MPT.	L D	1 1.5	
TP5	1. Publications, to include: a. Layout; b. Fault Isolation Procedures; c. Fault Codes; d. CU; and e. UPS;	L	0.5	

PO 001 – Maintain the Multi Ammunition Softkill System

EO 001.02 Initialize the Multi Ammunition Softkill System

1. Performance. Initialize the MASS.
2. Conditions:
    a. Given:
        (1) supervision;
        (2) references;
        (3) DWAN;
        (4) tools and test equipment;
        (5) MASS and/or MPT
        (6) PPE; and
        (7) assistance.
    b. Denied:
        (1) Nil.
    c. Environmental:
        (1) any shipboard conditions at sea or alongside; and
        (2) day or night
3. Standard. The Trainee shall initialize the MASS, IAW refs, by:
    a. observing all safety precautions;
    b. powering up/down;
    c. conduct system functional checks;
    d. verify operational status;
    e. verify operational readiness; and,
    f. report system status to supervisors.
4. Time Required:
    a. 7 x 50 min periods.
5. Instructional Methods:
    a. lecture;
    b. demonstration; and
    c. Trainee practice.
6. References:
    a. A1 – A7; and

b. E1 – E10.

7. Training Aids:

a. MASS System and/or MPT;

b. audio visual aids;

c. references; and

d. handouts.

8. Test Details:

a. EO 001.02 will be tested as part of PC 001 IAW Annex K.

9. Remarks:

a. Projects and exercises may be assigned;

b. MPT may be utilized; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference
TP1	1. Initialize the MASS, to include; a. power up. b. conduct system functional checks. c. verify operational status. d. verify operational readiness. e. report system status to supervisor f. power down	L D P	1 2 4	

PO 001 – Maintain the Multi Ammunition Softkill System

EO 001.03 Operate the Multi Ammunition Softkill System

1. Performance. Operate the MASS.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) MASS and/or MPT

      (6) PPE; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall operate the MASS, IAW refs, by:

   a. initializing;

   b. localizing fault to LRU;

   c. repairing system;

   d. verifying operational readiness;

   e. report system status to supervisor; and

   f. powering down.

4. Time Required:

a. EO – 51 x 50 min periods;

b. EC - 1 X 50 min periods; and

c. PC - 18 X 50 min periods.

5. Instructional Methods:

a. lecture;

b. Demonstration;

c. practice; and

d. Ships tour.

6. References:

a. A1 – A7; and

b. E1 – E10.

7. Training Aids:

a. MASS System and/or MPT;

b. audio visual aids;

c. references; and

d. Ship Tour.

8. Test Details:

a. EO 001.03 will be tested with EC 001.03 and as part of PC 001.

b. EC 001.03 will be a written test with a 1 period time limit IAW Annex K.

c. PC 001 and debrief will be conducted IAW Annex K with a 90 min time limit. (Corrective measures will start at minute 85).

9. Remarks:

a. Projects and exercises may be assigned;

b. MPT may be utilized;

c. Assignments are to be completed during non-instructional time;

d. 15 periods for practice PC to be conducted IAW Annex L; and

e. 3 Periods for PC pre/post brief and fault reinsertion.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference
TP1	1. Maintain the MASS system a. initializing; b. localizing fault to LRU; c. repairing system d. verify operational readiness. e. report system status to supervisor f. Power down	L D P	1 2 44	
TP2	1. Ships tour of UPS (UPS not Available at NFS(P))	tour	4	

PO 001 – Maintain the Multi Ammunition Softkill System

EO 001.04 Perform Pre/Post Firing Check on the Multi Ammunition Softkill System

1. Performance. Perform Pre/Post Firing check on the MASS.
2. Conditions:
    a. Given:
        (1) supervision;
        (2) references;
        (3) DWAN;
        (4) tools and test equipment;
        (5) MASS and/or MPT
        (6) PPE; and
        (7) assistance.
    b. Denied:
        (1) Nil.
    c. Environmental:
        (1) any shipboard conditions at sea or alongside; and
        (2) day or night.
3. Standard. The Trainee shall perform pre / post firing check on the MASS, IAW refs, by:
    a. performing Pre-firing routine; and
    b. performing Post-firing routine.
4. Time Required:
    a. EO = 4 x 50 min periods.
5. Instructional Methods:
    a. demonstration; and
    b. Trainee practice.
6. References:
    a. A1 – A7; and
    b. E1 – E10.
7. Training Aids:
    a. MASS System and/or MPT;
    b. audio visual aids;
    c. references; and

d. handouts.

8. Test Details:

a. n/a.

9. Remarks:

a. MPT may be utilized.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference
TP1	1. Perform pre-firing check on the MASS, to include: a. Ensure System is VETO; b. Verify UPS breakers; c. Verify Mode Selection and Operational Key switch position (both MP), to include: (1) Mode- REMOTE; (2) Operate- OPERATE; and (3) Emergency Stop depressed. d. verify all breakers inside both CU's; to include: (1) switch on CU; and (2) Verify CU's achieve	D P	.5 1.5	

	REMOTE e. power on MIU; f. power on MFD(s); g. ensure all cells are empty to include: (1) Lau State- REMOTE green (2) Emerg Stop green (3) Lau Pos 270, 0 and 90, 0 (4) VETO red; and (5) Return to Main page h. MP- ENBL light extinguished, to include: (1) Perform lamp test at MP- Verify all lamps illuminate; (2) Mode key switch to LOAD position; (3) Verify SAFE lamp illuminates; and (4) Mode key switch to LOCAL position Launcher to Load Position. i. Press and hold UP button till Launcher reaches load position. j. Mode selection key to LOAD remove key and verify safe lamp illuminates. k. Load launcher(s) with OMNI TRAP ammunition, to include: (1) Min 12 OMNI TRAP; (2) Ensure Launcher is clear of all material befor			

	leaving area; and (3) Ensure sentry is posted. l. Turn Mode selection key to REMOTE and ensure that the REMOTE light starts to blink and launcher returns to base position m. Enable MASS system; n. MFD ensure Enable green, to include; (1) all other indications green; and (2) press EMERGENCY Takeover o. THREAT page move threats from TDF to the TOB and apply; p. MFD Status page, to include (1) verify cells load with OMNITRAP and DUERAS ammunition; and (2) Enable green q. MAIN page, to include: (1) select spot on polar view; (2) ok threat (Port or Starboard); and (3) Verify threat on Polar view and Situation data sections. r. Select ID in situation data section, to include: (1) verify polar and threat ID view bolds (2) verify arrow beside FIRE button illuminates press FIRE			

	(3) ensure engagement status changes to executed (4) If Load State is selected for MAIN page verify subtraction of ammunition or return to Status page and verify cell(s) have fired; and (5) MAIN page select Long Range button and green area highlights on polar view. s. Select position within green area and select ok, to include: (1) verify in situation data section- Long Range Eng (2) select ID verify that polar and Situation data bold (3) verify flashing arrow beside fire button t. press fire u. engagement status – successfully fired is shown v. verify that Load State display correct ammunition count or verify on w. status page state of launcher x. repeat steps 9- 21 for second launcher; y. VETO MASS system, to include (1) Verify at MFD Main page VETO red; and (2) all other indications green z. Place sentry at launcher to			

	be moved; aa. MP- Mode selection key to LOCAL, to include bb. Move launcher to load position, by pressing UP button; cc. Mode selection key to LOAD and verify Safe Lamp illuminates; and dd. Remove key ee. Unload all ammunition from launcher, to include: (1) Ensure area around launcher is clear; and, (2) sentry is posted. ff. MP- set Mode selection key to REMOTE, to include gg. verify that REMOTE light blinks; and. hh. launcher returns to base position ii. MFD verify all indications are correct, to include; jj. VETO red; and kk. Status page all launcher cells cleared. ll. Set launcher to required state and report to CoC state of MASS system.			
TP2	1. Perform post firing check on the MASS: 2. Post firing check to be conducted IAW C-58-647-000/NY-Z01 01 post firing	D P	.5 1.5	

PO 002 – Maintain Ship's Main Gun System

EO 002.01 Describe the Borfors 57mm Gun System

1. Performance. Describe the Bofors 57mm Gun System.
2. Conditions:
   a. Given:
      (1) supervision;
      (2) references;
      (3) DWAN;
      (4) tools and test equipment;
      (5) PPE;
      (6) Main Gun System; and
      (7) assistance.
   b. Denied:
      (1) Nil
   c. Environmental:
      (1) any shipboard conditions at sea or alongside; and
      (2) day or night.
3. Standard. The Trainee shall describe the Bofors 57mm Gun System, IAW refs, by:
   a. capabilities and limitations;
   b. purpose;
   c. components;
   d. component layout;
   e. power distribution;
   f. interfaces;
   g. ammunition types; and
   h. mount safety and operation.
4. Time Required:
   a. EO = 5 x 50 min periods; and
   b. EC = 1 X 50 min period.
5. Instructional Methods:
   a. lecture; and
   b. demonstration.
6. References:

a. A1 – A7; and

b. B1 – B12.

7. Training Aids:

a. Bofors 57mm Gun System and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 002.01 will be tested with EC 002.01 and PC 002 IAW Annex K; and

b. 1 period for EC 002.01 written test IAW Annex K.

9. Remarks:

a. MPT may be utilized,

b. Projects and exercises are to be completed during non-instructional time

10. Teaching Points;

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference
TP1	1. Describe the 57mm gun system, to include: a. capabilities and limitations; b. purpose; c. firing system; d. mount ammo capacity/	L	0.5	
	configuration; e. reload times; f. ranges and accuracy; and g. laying system stabilization			

	2. components to include: 3. a. Platform; b. Ammo Hoists; c. Gun Cupola; d. Elevating System; e. Loading System; f. Hydraulic System; g. Electrical System; h. Laying System; and i. Control Logic System Component layout	L D & Tour of System	0.5 1	
	4. Power distribution; and 5. Ship's power	L	0.5	
	6. WIC; STIRs; and 7. ammunition types 8.	L	0.5	
TP2	1. Describe the 57mm gun system mount safety and operation, to include: 2. Safety Hazards; and 3. Safety Implements	L D	1 1	

PO 002 – Maintain Ship's Main Gun System

EO 002.02 Perform Firing Checks on the Bofors 57mm Gun System

1. Performance. Perform firing checks on the Bofors 57mm Gun System.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Main Gun System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall perform firing checks on the Bofors 57mm Gun System, IAW refs, by:

   a. explaining INO requirements;

   b. conducting pre firing check; and

   c. conducting post firing check.

4. Time Required:

   a. EO – 14 x 50 min periods.

5. Instructional Methods:

   a. lecture; and

   b. Trainee practice.

6. References.

a. A1 – A7; and

b. B1 – B12.

7. Training Aids:

a. Bofors 57mm Gun System;

b. audio visual aids;

c. handouts; and

d. references.

8. Test Details:

a. EO 002.02 will be tested with PC 002 IAW Annex K

9. Remarks:

a. MPT may be utilized; and

b. Projects and exercises are to be completed during noninstructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference
TP1	1. Explain INO requirements	L	1	
TP2	1. Perform pre firing routine	P	6.5	
TP3	1. Perform post firing routine	P	6.5	

PO 002 – Maintain Ship's Main Gun System

EO 002.03 Operate the Bofors 57mm Gun Laying Unit / Control Panel

1. Performance. Operate Bofors 57mm Gun Laying Unit / Control Panel.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Main Gun System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall operate the Bofors 57 mm Gun Laying Unit / Gun Control Panel, by:

   a. describing Gun Laying Unit /Gun Control Panel; and

   b. conducting Bofors 57 mm Gun Laying Unit / Gun Control Panel operations, to include:

      (1) powering up, powering down;

      (2) carrying out equipment functional check(s);

      (3) verifying status;

      (4) verifying operational readiness; and (5) reporting equipment status to supervisor.

4. Time Required:

a. EO – 15 x 50 min periods

b. EC - 1 X 50 min period

5. Instructional Methods:

a. lecture;

b. demonstration;

c. Trainee practice.

6. References:

a. A1 – A7; and

b. B1 – B12.

7. Training Aids:

a. Bofors 57mm Gun System and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 002.03 will be tested with EC 002.03 and PC 002 IAW Annex K; and

b. 1 period for EC 002.03 written test IAW Annex K.

9. Remarks:

a. MPT may be utilized; and

b. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference

TP1	1. Describe the Gun Control Panel(GCP), to include: a. Operation of GCP; b. Alert Levels, to include; (1) Power Off; (2) Power Alert; (3) Green; (4) Red: (5) Misfire (6) Service; (7) Test; and (8) Emergency Stop; 2. Control Modes, to include: a. Remote Control; b. Local Control 1; and c. Local Control2 m Function for 3. System Alert Levels, to include: a. GCP Display; b. Basic Display; c. Function Key Display; d. Combat Display; e. Information Display; f. Loading Display; g. Setup Display; h. Test Display; and i. Special Display.	L D P	2 2 8	

TP2	1. Describe the Gun Laying Unit to include: Components to include: a. Emergency stop; b. Fire; c. Blocking; d. Power On/Off; e. Unsafe; f. Target Tracking (TRACK); g. Laying Joystick; and h. Hand Support; i. Overview.	L D	1.5 1.5	

PO 002 – Maintain Ship's Main Gun System

EO 002.04 Perform Bofors 57mm Gun Close Up Clear Away / Misfire Procedures

1. Performance. Perform Bofors 57mm Gun Close Up Clear Away / Misfire Procedures.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Main Gun System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall perform Bofors 57mm Gun Close up Clear Away / Misfire procedures, IAW refs, by:

   a. conducting Bofors 57mm Gun Close up Clear Away / Misfire procedure functions, to include:

      (1) initializing;

      (2) localizing fault to LRU;

      (3) repairing and aligning system;

      (4) (4) verifying operational readiness; and

      (5) powering down.

4. Time Required:

   a. EO – 48 x 50 min periods; and

b. EC - 1 X 50 min period

5. Instructional Methods:

a. lecture;

b. demonstration; and

c. Trainee practice.

6. References:

a. A1 – A7; and

b. B1 – B12.

7. Training Aids:

a. Bofors 57mm Gun System and/or MPT;

b. audio visual aids;

c. handouts; and

d. references.

8. Test Details:

a. EO 002.04 will be tested with EC 002.04 and as part of PC 002; and.

b. EC 002.04 will be a written test with a 1 period time limit, IAW Annex K

9. Remarks:

a. MPT may be utilized; and

b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference

TP1	1. Perform Close Up Clear Away and Misfire Procedure, to include: a. Close up Clear away Procedure; (1) Foreign Object (2) Debris; (FOD) b. Open Breech; (1) Breech and Barrel Inspection (2) Cleaning of the Barrel (3) Closing Breech / Ease Rammer (4) Check Fluids (5) Pre-firing Checks (6) Automatic Test (7) Loading c. Misfire Procedure; (1) Definition: (a) Hot Gun (b) Hang Fire (c) Cookoff d. Check Loading; e. Firing Stop; f. Misfire; g. Removal of round; h. Checking Fluids; i. Pre-firing Checks; j. Automatic Test;	L D P	2 8 38	

PO 002 – Maintain Ship's Main Gun System

EO 002.05 Maintain the Bofors 57mm Gun Power Distribution System

1. Performance. Maintain the Bofors 57mm Gun power distribution system

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Main Gun System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the Bofors 57mm Gun power distribution system, IAW refs, by:

   a. describing the power distribution system; and

   b. conducting Bofors 57mm Gun power distribution system functions, to include:

   c. powering up, powering down;

   d. carrying out equipment functional check(s);

   e. verifying status;

   f. verifying operational readiness; and

   g. reporting equipment status to supervisor.

4. Time Required:

a. EO – 20 x 50 min periods

b. EC - 1 X 50 min period

5. Instructional Methods:

a. lecture;

b. demonstration; and

c. Trainee practice.

6. References:

a. A1 – A7

b. B1 – B12.

7. Training Aids:

a. Bofors 57mm Gun System and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 002.05 will be tested with EC 002.05 and PC 002, IAW Annex K; and

b. 1 period for EC 002.05 written test IAW Annex K.

9. Remarks:

a. MPT may be utilized; and

b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference

TP1	1. Describe Power to include: a. Below Deck; b. Distribution box A52; c. Rectifier A14; d. Transformer T11; e. Battery G11; f. Connecting Panel and g. Filter Z11. 2. Upper Mounting to include: a. Slipring A50; and b. Gun Control Logic A11; c. Power Distribution Box A12; and d. Sensors. 3. Circuit Diagrams to include: a. 440 VAC 3Φ 60Hz; b. 230 VAC 3Φ 60Hz; c. 28VAC 400Hz reference voltage; d. 28 VDC Standby; and e. 28 VDC 70 Amps.	L	5	
TP2	1. Describe the power-on/ power-off sequence and initial conditions to include: a. indications; and b. sequence of events:	L D	1 2	

TP3	1. Localize fault to LRU to include: a. symptom recognition; b. use of fault finding logic charts; c. appropriate test points; d. logical procedure; e. interpretation of schematics; f. Repair system; g. tests and adjustments; and h. verify operational readiness.	P	12	

PO 002 – Maintain Ship's Main Gun System

EO 002.06 Describe the Operation of the Bofors 57mm Gun System

1. Performance. Describe the operation of the Bofors 57mm Gun System

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Main Gun System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall describe the operation of the Bofors 57mm Gun hydraulic system, IAW refs, by:

   a. explaining designators; and

   b. explaining Bofors 57 mm Gun hydraulic subsystems.

4. Time Required:

   a. EO =10 x 50 min periods; and

   b. EC=1 x 50 min periods.

5. Instructional Methods:

   a. lecture;

   b. demonstration; and

c. Trainee practice.

6. References:

a. A1 – A7; and

b. B1 – B12.

7. Training Aids:

a. Bofors 57mm Gun System and/or MPT;

b. audio visual aids;

c. handouts; and

d. references.

8. Test Details:

a. EO 002.06 will be tested with EC 002.06 and PC 002, IAW Annex K; and

b. 1 period for EC 002.06 written test IAW Annex K.

9. Remarks:

a. MPT may be utilized; and

b. assignments with block diagrams of hydraulic system shall be given, and are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference

TP1	1. Describe the operation of the hydraulic system, to include: a. Designators; b. 2 Hydraulic Subsystems; c. Oil Supply 10/20 Bar; d. Oil Supply 100 bar; e. Laying System; f. Locking Plungers; g. Automatic Load System; 2. Elevating (Firing)System; and 3. Oil Cooling System	L D P	4 2 4	

PO 002 – Maintain Ship's Main Gun System

EO 002.07 Maintain the Bofors 57mm Gun Loading/Firing System

1. Performance. Maintain the Bofors 57mm Gun Loading/Firing System.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Main Gun System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the Bofors 57mm Gun Loading/Firing System, IAW refs, by:

   a. explaining the Bofors 57 mm Loading and Firing system; and

   b. conducting Bofors 57 mm Loading and Firing system functions, to include:

      (1) initializing;

      (2) localizing fault to LRU;

      (3) repairing and aligning system; (4) verifying operational readiness; and

      (5) powering down.

4. Time Required:

   a. EO – 35 x 50 min periods; and

   b. EC –1 x 50 min periods

5. Instructional Methods:

   a. lecture;

   b. demonstration; and

   c. Trainee practice.

6. References:

   a. A1 – A7; and

   b. B1 – B12.

7. Training Aids:

   a. Bofors 57mm Gun System and/or MPT;

   b. audio visual aids;

   c. handouts; and

   d. references.

8. Test Details:

   a. EO 002.07 will be tested with EC 002.07 and PC 002, IAW Annex K; and

   b. 1 period for EC 002.07 written test IAW Annex K.

9. Remarks:

a. MPT may be utilized; and

b. Assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference
TP1	1. Describe the 57mm loading system, to include:	L	4	
	a. Hoist;	D	4	
	b. Round receiver;			
	c. Loading cassette;	P	11	
	d. Cassette positioning			
	e. Intermediate magazine;			
	f. Gun magazine;			
	g. Lateral feed; and			
	h. Feed ruler components;			
	2. Firing system components to include:			
	a. Rammer;			
	b. Breech housing;			
	c. Recoil mechanism; and			
	d. Case ejection.			
	3. Ramming and firing circuit sequence to include:			
	a. Mechanical functions; and			
	b. Electrical functions			

TP2	1. Localize fault to LRU to include: a. symptom recognition; b. use of fault finding logic charts; c. appropriate test points; d. logical procedure; e. interpretation of schematics; f. Repair and align system; g. Tests and adjustments; and h. Verify operational readiness	P	16	

PO 002 – Maintain Ship's Main Gun System

EO 002.08 Maintain the Bofors 57mm Gun Laying System

1. Performance. Maintain the Bofors 57mm Gun laying System.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Main Gun System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

(1) any shipboard conditions at sea or alongside; and

(2) day or night.

3. Standard. The Trainee shall maintain the Bofors 57mm Gun laying System, IAW refs, by:

a. describing the Bofors 57mm Gun Laying system; and

b. conducting Bofors 57mm Gun Laying system functions, to include:

(1) initializing;

(2) localizing fault to LRU;

(3) repairing and aligning system; (4) verifying operational readiness; and

(5) powering down.

4. Time Required:

a. EO – 31 x 50 min periods; and

b. EC – 1 x 50 min periods.

5. Instructional Methods:

a. Lecture;

b. Demonstration; and

c. Trainee practice.

6. References:

a. A1 – A7; and

b. B1 – B12.

7. Training Aids:

a. Bofors 57mm Gun System and/or MPT;

b. audio/visual aids;

c. handouts; and

d. references.

8. Test Details:

a. EO 002.08 will be tested with EC 002.08 and PC 002, IAW Annex K; and

b. 1 period for EC 002.08 written test IAW Annex K.

9. Remarks:

a. MPT may be utilized; and

b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference
TP1	1. Describe the 57mm laying system, to include: a. Gun Control Computer (GCC) : b. Servo Amplifier to include: (1) General: (2) Modes of Control: and (3) Modes of Operation. c. Aiming and Fire Limitations (AFL) to include; (1) General; (2) Aiming Limitation; and (3) Fire Limitation.	L D	1 1	
	2. Laying System Elevation and Training, functional description and signal flow, to include: a. Servo motor; b. Force motor; c. Digital Resolver; d. Hydraulic Gearing; and e. gyro, functional description and block diagram signal flow.	L	1	

TP2	1. Localize fault to LRU by utilizing: a. symptom recognition; b. use of fault finding logic charts; c. appropriate test points; d. logical procedure; e. interpretation of schematics; f. Repair and align system; g. Tests and adjustments; and h. Verify operational readiness.	P	28	

PO 002 – Maintain Ship's Main Gun System

EO 002.09 Maintain the Bofors 57mm Gun Control Computer (GCC) and Gun Control Logic Box (GCL)

1. Performance. Maintain the Bofors 57mm GCC and GCL.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) Main Gun System; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the Bofors 57mm GCC and GCL, IAW refs, by:

   a. explaining the GCC and GCL;

   b. conducting GCC and GCL functions, to include:

      (1) initializing;

      (2) localizing fault to LRU;

      (3) repairing and aligning system; (4) verifying operational readiness; and

      (5) powering down.

4. Time Required:

   a. EO – 20 x 50 min periods;

   b. EC – 1 x 50 min periods; and

   c. PC – 17 X 50 Min Periods.

5. Instructional Methods:

   a. Lecture;

   b. Demonstration; and

   c. Trainee practice.

6. References:

   a. A1 – A7; and

   b. B1 – B12.

7. Training Aids:

   a. Bofors 57mm Gun System and/or MPT;

   b. handouts;

   c. laptop diagnostic computer;

   d. audio/visual aids; and

   e. references.

8. Test Details:

   a. EO 002.09 will be tested with EC 002.09 and PC 002 IAW Annex K,

b. 1 period for EC 002.09 written test IAW Annex K; and

c. PC 002 and debrief to be conducted IAW Annex K with a 90 min time limit (Corrective measures start at minute 85).

9. Remarks:

a. MPT may be utilized; and

b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference
P1	1. Describe the 57mm GCC, to include:	L	2	
	a. Describe function of the GCC to include:	D	2	
	(1) Overview (2) Indicators and Connection ; (3) Functions; (4) Basic Laying; (5) Aiming / Firing Limitations; (6) Built in Test (BIT); (7) Start Test; (8) Continuous Test; and (9) Automatic Test	P	16	
	b. Data Recording:			
	c. Gun Main Tester (GMT) to include:			
	(1) Events Log;			

	(2) Laying Data Log; (3) Gun Control Logic Box (GCL) (4) Overview and Construction (5) Upper GCL (6) Circuit Card Assembly (7) Double Sided Relay Board; (8) FIOC Board (A110); (9) Power Unit			

PO 003 – Maintain Close In Weapons System (CIWS)

EO 003.01 Describe the CIWS

1. Performance. Describe the CIWS.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall describe the CIWS, IAW refs, by:

   a. purpose;

   b. capabilities and limitations;

   c. components;

d. component layout;

e. power distribution;

f. interfaces; and

g. ammunition.

4. Time Required:

a. EO - 6 x 50 min periods.

5. Instructional Methods:

a. lecture; and

b. demonstration.

6. References:

a. A1 – A7, and

b. C1 – C20.

7. Training Aids:

a. CIWS System and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 003.01 will be tested with EC 003.10 and PC 003, IAW Annex K.

9. Remarks:

a. Projects and exercises may be assigned; and

b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference

TP1	1. Describe the CIWS, to include: a. capabilities and limitations: b. purpose; c. rate of fire; d. firing modes; e. ranges and accuracy; f. mount ammo capacity; g. reload times; h. components to include:;	L	4	
	(1) mount (2) radome (3) gun body (4) barbette (5) ELX (6) RCS, (7) LCS (8) LCP; and (9) salt water cooling			
TP2	1. component layout;	Demo/Tour of system	1	
TP3	2. power distribution: a. ship's supply; b. interfaces; and c. ammunition types	L	1	

PO 003 Maintain Close In Weaspons System (CIWS)

EO 003.02 Initialize the CIWS

1. Performance. Initialize the CIWS.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall initialize the CIWS, IAW refs, by:

a. conducting CIWS Daily 1 (D1s) maintenance routines;

b. CIWS loading /unloading; and

c. securing CIWS to the Battery Off state.

4. Time Required:

a. EO - 8 x 50 min periods.

5. Instructional Methods:

a. demonstration; and

b. Trainee practice.

6. References:

a. A1 – A7, and

b. C1 – C20.

7. Training Aids:

a. CIWS System and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 003.02 will be tested with EC 003.10 and PC 003, IAW Annex K.

9. Remarks:

a. MPT may be utilized,

b. Projects and exercises may be assigned;

c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference

TP1	1. Initialize CIWS, to include: a. Close up/clear away; b. carrying out D1s; c. load operational program; and d. Operate Windows Parameter Analysis Storage System (BRICKPASS)	D P	0.5 3	
	2. loading	P	2	
	3. Unloading	P	2	
	4. securing CIWS to the Battery Off state	D	0.5	

PO 003 Maintain Close In Weaspons System (CIWS)

EO 003.03 Describe the CIWS Peculiar Support Equipment (PSE)

1. Performance. Describe the CIWS PSE.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS; and

      (7) assistance.

   b. Denied:

      (1) Nil.

c. Environmental:

(1) any shipboard conditions at sea or alongside; and

(2) day or night.

3. Standard. The Trainee shall describe the CIWS PSE, IAW refs, by:

a. describing the CIWS PSE components.

4. Time Required:

a. EO - 7 x 50 min periods

5. Instructional Methods:

a. lecture; and

b. demonstration;

6. References:

a. A1 – A7, and

b. C1 – C20.

7. Training Aids:

a. CIWS System and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. N/A.

9. Remarks:

a. Projects and exercises may be assigned; and

b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference

TP1	1. Describe CIWS PSE, to include: a. aft gun mount assembly; b. transport handles; c. alignment pins; d. vented desiccators; e. davit assembly; f. ice protection bag; g. telescope lens cover; h. lifting sling; i. purge adapter; j. portable maintenance unit;	L D	3 4	

		k. thermal imager test cable assembly; l. STAB carrying handle; m. anti-rotation device; n. barrel-spreader bars; o. test point interconnection box; p. card extenders; q. component removal extension set; r. gun extension set; s. maintenance platform; and t. Radar Servo Maintenance Unit (RSMU).			

PO 003 Maintain Close In Weaspons System (CIWS)

EO 003.04 Maintain the CIWS Power Supply and Control Group Subsystem

1. Performance. Maintain the CIWS Power Supply and Control Group subsystem (PSCG).

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the CIWS PSCG Subsystem, IAW refs, by:

   a. describing the CIWS PSCG subsystem; and

   b. conducting fault finding to the CIWS PSCG subsystem's LRU.

4. Time Required:

   a. EO - 22 x 50 min periods.

5. Instructional Methods:

   a. lecture;

   b. demonstration; and

   c. Trainee practice.

6. References:

   a. A1 – A7, and

   b. C1 – C20.

7. Training Aids:

   a. CIWS System and/or MPT;

   b. references;

   c. audio visual aids; and

   d. handouts.

8. Test Details:

   a. EO 003.04 will be tested with EC 003.10 and PC 003, IAW Annex K.

9. Remarks:

   a. Projects and exercises may be assigned; and

   b. assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference

TP1	1. Describe the CIWS PSCG subsystem, to include: a. subsystem overview; b. Power Supply and Control Group (PSCG): c. Power Quality Monitor (PQM): d. battle short circuit. e. ELX and EM power supplies; f. 115 VAC power generation;	L D	10 4	

	g. 400 Hz power generation; h. Power control relays and circuit breakers; and i. BITE circuit cards.			

	2. subsystem diagrams to include: a. PSCG ; b. LCP/RCP; c. 115V 60 Hz; d. 440V 60Hz; e. +/- 55VDC; f. +28VDC; g. 115V and 26V 400Hz; h. +/- 29VDC; i. +/- 28.5VDC; j. +/- 8.5VDC; k. EM power supply; and l. +27 VDC.			
	3. Describe mode sequencing to include: a. battery off mode; b. standby; c. standby maintenance submode; d. standby ELX; e. air ready; f. air ready maintenance; g. AAW Manual;			

	h. AAW Auto; i. Surface; and j. Battle Short			
TP2	1. Localize fault to LRU to include: a. symptom recognition; b. use of fault finding logic charts; c. appropriate test points; d. logical procedure; e. interpretation of schematics; f. Repair and align system; g. Tests and adjustments; and h. Verify operational readiness	P	8	

PO 003 Maintain Close In Weaspons System (CIWS)

EO 003.05 Maintain the CIWS Environmental Control Group (ECG) Subsystem

1. Performance. Maintain the CIWS ECG Subsystem.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS; and

      (7) assistance.

   b. Denied:

      (1) Nil

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the CIWS ECG Subsystem; by:

   a. describing the CIWS ECG subsystem; and

   b. conducting CIWS ECG subsystem functions, to include:

      (1) initializing;

(2) localizing fault to LRU;

(3) repairing and aligning system;

(4) verifying operational readiness; and

(5) powering down.

4. Time Required:

a. EO - 22 x 50 min periods.

5. Instructional Methods:

a. lecture; and

b. demonstration; and

c. trainee practice.

6. References:

a. A1 – A7, and

b. C1 – C20.

7. Training Aids:

a. CIWS System and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 003.05 will be tested with EC 003.10 and PC 003 IAW Annex K.

9. Remarks:

a. Projects and exercises may be assigned; and

b. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference

TP1	1. Describe the CIWS ECG, to include: a. functional description; b. power supply and distribution;	L D	10 4	

	c. motor power control circuits. d. electronic power supply circuits. e. coolant control and monitoring functions; and f. coolant pump and control; 2. Alternate coolant pump activation circuits to include: a. coolant temperature monitoring; b. coolant flow rate monitoring; c. coolant fluid circuits (schematic). d. air control and monitoring functions. e. Blower and compressor motor control circuits; f. Air sensor and indicator logic circuits; g. Radome heater and fan unit control, sensor and indicators; and h. Cooling air circuit and waveguide dry air supply (schematic). 3. Identify fault			

	indications to include: a. Local Control Panel (LCP) lamps; b. ECG electronic assembly front panel lamps; and c. ECG flow meters and gauges.			
TP2	1. Localize fault to LRU to include: a. symptom recognition; b. use of fault finding logic charts; c. appropriate test points; d. logical procedure; e. interpretation of schematics; f. Repair and align system; g. Tests and adjustments; and h. Verify operational readiness	P	8	

PO 003 Maintain Close In Weaspons System (CIWS)

EO 003.06 Maintain the CIWS Built In Test Equipment (BITE)

1. Performance. Maintain the CIWS BITE.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the CIWS BITE, IAW refs, by:

   a. describing the CIWS BITE; and

   b. conducting CIWS BITE functions, to include:

      (1) initializing;

      (2) localizing fault to LRU;

      (3) repairing and aligning system; (4) verifying operational readiness; and

      (5) powering down.

4. Time Required:

   a. EO - 18 x 50 min periods.

5. Instructional Methods:

   a. lecture; and

   b. Trainee practice.

6. References:

   a. A1 – A7, and

   b. C1 – C20.

7. Training Aids:

   a. CIWS System and/or MPT;

   b. references;

   c. audio visual aids; and

   d. handouts.

8. Test Details:

   a. EO 003.06 will be tested with EC 003.10 and PC 003 IAW Annex K.

9. Remarks;

   a. Projects and exercises may be assigned;

   b. MPT may be utilized; and

   c. assignments are to be completed during non-instructional time.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference
TP1	1. Describe the CIWS BITE Subsystem to include: a. Test Point Selector Unit (TPSU) BITE Control and Interface: b. Test Interface Modules	L	10	

	(TIM) function; c. TPSU measurement select control word W1 function; d. TPSU measurement setups; e. TPSU TIM control word W2; f. TPSU stimulus control word W3; g. TPSU measurement initiate control word W4 function; h. TPSU data request control word W5 and TCU data output function; i. TPSU clear all TIM functions to include: j. TPSU to TCU loop test function; and k. TPSU/TCU loop interface function; and l. Computer BIT, Fault Isolation Test (FIT), initialization and SOT 1;			

	2. LCP/WCC BITE control and interface to include: a. LCP to WCC test control and interface function; b. WCC to TCU data and address inputs; c. WCC to TCU measurement select control word W1 data transfer; d. WCC to TCU/TIM stimulus input control word W3 data transfer;			
	e. WCC measurement initiate control word W4; f. frequency measurement; g. voltage measurement; h. WCC data request word W5 and TCU to WCC data transfer of test results to include: (1) BITE self test circuits; and (2) TCU/TIM wrap around SOT 2.			

TP2	3. Localize fault to LRU to include: a. symptom recognition; b. use of fault finding logic charts; c. appropriate test points; d. logical procedure; e. interpretation of schematics; f. Repair and align system; g. Tests and adjustments; and h. Verify operational readiness	P	8	

PO 003 Maintain Close In Weaspons System (CIWS)

EO 003.07 Maintain the CIWS Weapon Control Computer (WCC)

1. Performance. Maintain the CIWS WCC.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

(6) CIWS; and

(7) assistance.

b. Denied:

(1) Nil.

c. Environmental:

(1) any shipboard conditions at sea or alongside; and

(2) day or night

3. Standard. The Trainee shall maintain the CIWS WCC, IAW refs, by:

a. describing the CIWS WCC; and

b. conducting CIWS WCC functions, to include:

(1) initializing;

(2) localizing fault to LRU;

(3) repairing and aligning system; (4) verifying operational readiness; and

(5) powering down.

4. Time Required:

a. EO - 17 x 50 min periods.

5. Instructional Methods:

a. lecture;

b. demonstration; and

c. trainee practice.

6. References:

b. A1 – A7, and

c. C1 – C20.

7. Training Aid:

a. CIWS System and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 003.07 will be tested with EC 003.10 and PC 003 IAW Annex K.

9. Remarks:

a. Projects and exercises may be assigned;

b. MPT may be utilized; and

c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference
TP1	1. Describe the CIWS WCC to include; a. General Circuit Description: b. Description Of The Weapon Control Group Subsystem (WCC); and c. High Order Language Computer (HOLC) including	L	8	

	activation and subsystem control Circuitry; 2. Interrupt and data input Circuitry to include: a. Data Output Circuitry; b. Timing Control Circuitry; c. Antenna Position and Radiate Control Circuitry; d. Mount Status and Analog-toDigital Conversion Circuitry; e. Mount Data Output and Digitalto-Analog Conversion Circuitry; f. WCC to Signal Processor Interface Circuitry; g. Computer Test Circuitry; and h. WCC / LCP / PSE Loop Circuitry.			

	3. Functional Circuit Theory to include: a. Voltage Sensing; b. Computer Input and Output Control; c. Input and Output command decoding and distribution; d. Computer Interrupt Function; e. Data Input Function;			

	f. Computer timing control; g. Antenna Position and Radiate Control; h. Antenna switching; i. RADZONE determination; j. Mount Status and A/D Conversion Function; k. Mount Output and D/A Conversion Function; l. Digital to Analog Converter; m. Computer Operational Program; n. Computer Data Output Function; o. WCC – Signal Processor Interface; p. Computer Test Function; q. WCC – LCP – PSE Loop; r. Data Input From the DATA Terminal; and s. Program installation from BRICKPASS to MSDD Hard Disk.			

TP2	1. Localize fault to LRU to include: a. Review WCC section of Vol 5 fault isolation; b. Review all SOTs and FITs that pertain to the WCC subsystem; c. symptom recognition; d. use of fault finding logic charts; e. appropriate test points; f. logical procedure; g. interpretation of schematics; h. Repair and align system; i. Tests and adjustments; and j. Verify operational readiness.	D P	1 8	

PO 003 Maintain Close In Weaspons System (CIWS)

EO 003.08 Maintain the CIWS Operator Control and Display (C&D) Subsystem

1. Performance. Maintain the CIWS Operator C&D subsystem.
2. Conditions:
   a. Given:
      (1) supervision;
      (2) references;
      (3) DWAN;
      (4) tools and test equipment;
      (5) PPE;
      (6) CIWS; and
      (7) assistance.
   b. Denied:
      (1) Nil.
   c. Environmental:
      (1) any shipboard conditions at sea or alongside; and
      (2) day or night.
3. Standard. The Trainee shall maintain the CIWS Operator C&D subsystem, IAW refs, by:
   a. describing the CIWS operator C&D subsystem; and
   b. conducting CIWS operator C&D subsystem functions, to include:
      (1) initializing;
      (2) localizing fault to LRU;
      (3) repairing and aligning system; (4) verifying operational readiness; and
      (5) powering down.
4. Time Required:
   a. EO - 12 x 50 min periods.
5. Instructional Methods:
   a. lecture; and
   b. Trainee practice.
6. References:

a. A1 – A7, and

b. C1 – C20.

7. Training Aids:

a. CIWS System and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 003.08 will be tested with EC 003.10 and PC 003 IAW Annex K.

9. Remarks:

a. Projects and exercises may be assigned;

b. MPT may be utilized; and

c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference
TP1	1. Describe the CIWS C&D Subsystem, to include: a. Local Control Panel (LCP) b. LCP internal component identification; c. block subsystem overview: d. LCP/WCC	L	4	

	comm loop; e. WCC/LCP comm loop; f. LCP/RCP comm loop; g. RCP/LCP comm loop; and h. Peripheral functions. 2. LCP power distribution and indicator test function to include: a. LCP power application; b. 1A2PS1; c. 1A1PS2; d. Power on clear circuit; e. Lamp and read out test circuits; f. Power supply adjustment and test circuits; and g. Power removal. 3. RCP to include: a. RCP internal component identification; b. RCP power distribution and indicator test function: c. RCP power application; d. 4A1PS1; e. 4A1PS2; f. 4A1PS3;			

	g. RCP power			

		on clear circuit; h. Lamp and read out test circuits; i. Power supply adjustment and test circuits; and j. Power removal			

PO 003 Maintain Close In Weaspons System (CIWS)

EO 003.09 Maintain the CIWS Gun Subassembly

1. Performance. Maintain the CIWS Gun Subassembly.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the CIWS Gun Subassembly, IAW refs, by:

   a. describing the CIWS Gun subassembly;

   b. describing the operation of the pneumatic drive and gun control;

   c. removing/ disassembling/ assembling/ replacing/ testing the 20mm gun, ammo drum and the ammo conveyor system; and

   d. conducting CIWS Gun subassembly functions, to include:

      (1) initializing;

      (2) localizing fault to LRU;

      (3) repairing and aligning system;

      (4) verifying operational readiness; and

(5) powering down.

4. Time Required:

a. EO - 64 x 50 min periods

5. Instructional Methods:

a. lecture; and

b. Trainee practice.

6. References:

a. A1 – A7, and

b. C1 – C20.

7. Training Aids:

a. CIWS System and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 003.09 will be tested with EC 003.10 and PC 003 IAW Annex K.

9. Remarks:

a. Projects and exercises may be assigned;

b. MPT may be utilized; and

c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference

TP1	1. Describe the operation of the CIWS Gun Subassembly, to include: a. Elevation gun cradle assembly; b. 20mm gun: c. Barrels and accessories; d. Breech bolt assemblies;			
	e. Housing assembly and associated parts; and f. Rotor assembly and associated parts. g. Ammunition drum assembly: h. Outer drum; i. Inner drum; j. Scoop disc assemblies; and k. Drum cover assemblies. l. Ammunition conveyor assembly: m. Exit unit; n. Feed ammunition chutes; o. Transfer unit; p. Feeder unit; q. Unload drive assembly; r. Ammunition element chutes; s. Entrance unit; and t. Loader assembly.	L	4	

	2. Describe the operation of the pneumatic drive and gun control, to include: a. Gun pneumatic drive function: b. Identify pneumatic components; c. Activation time delay; d. Clutch activation; e. Auto power submode; f. Gun firing cycle; g. Ammunition loading; and h. BITE interface.			
TP2	1. Remove/ disassemble/ assemble/ replace/ test the 20mm gun, ammo drum, ammo conveyor system IAW MRCs.	P	45 Groups of 4	
TP3	1. Describe the operation of the pneumatic drive and gun control, to include: a. Gun Control Unit (GCU) power distribution/ gun go function to include:	L	7	

	(1) Fire volts power supply; (2) Logic power supplies; (3) Solenoid and gun ready power supplies; (4) Safe/ arm circuit; (5) Safe/ arm display circuits; (6) Fire cutout switches; (7) Temporary no fire zones; (8) Trigger and fire volts arm circuit; (9) Fire volts relay supply; (10) Gun system go circuits; (11) Perform and verify PSOT 13; and (12) Veto. (13) Gun firing/ clearing function: (14) Target data and system display; (15) LCP and RCP gun control circuits:			

	(16) AAW auto; (17) AAW Manual; (18) surface; and (19) Pre-Action Calibration (PAC). b. Fire/ hold fire display circuits; c. Gun firing cycle to include: (1) interlocked trigger and gun control valve logic circuits; (2) firing command interlock circuit; (3) rounds counter loading; (4) magnetic pickup device; (5) trigger interrupt; (6) rounds counting; and (7) burst firing. (8) Gun clearing cycle: (9) loss of trigger input signal; (10) loss of last round trigger signal input; and (11) loss of firing cutout input. (12) Maintenance and testing: (13) operability testing; (14) BITE interface; and (15) gun maintenance and clear command switch.			

TP4	1. Localize fault to LRU to include: a. symptom recognition; b. use of fault finding logic charts; c. appropriate test points; d. logical procedure; e. interpretation of schematics; f. Repair and align system; g. Tests and adjustments; and h. Verify operational readiness	P	8	

PO 003 Maintain Close In Weaspons System (CIWS)

EO 003.10 Maintain the CIWS Mount Servo Subsystem

1. Performance. Maintain the CIWS Mount Servo System

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the CIWS Mount Servo System, IAW refs, by:

   a. describing the CIWS Mount Servo subsystem; and

   b. conducting CIWS Mount Servo subsystem functions, to include:

      (1) initializing;

      (2) localizing fault to LRU;

      (3) repairing and aligning system; (4) verifying operational readiness; and

      (5) powering down.

4. Time Required:

   a. EO - 28 x 50 min periods; and

   b. EC - 17X 50 min periods.

5. Instructional Methods:

a. lecture;

b. demonstration;

c. practice; and

d. PC.

6. References:

a. A1 – A7, and

b. C1 – C20.

7. Training Aids:

a. CIWS System and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 003.10 will be tested with EC 003.10 IAW Annex K; and

b. EC 003.10 and debrief to be conducted with a 90 min time limit. (corrective measures start at minute 85).

9. Remarks:

a. Projects and exercises may be assigned;

b. MPT may be utilized;

c. assignments are to be completed during non-instructional time;

d. 12 periods shall be allotted to overall system fault finding practice; and

e. 2 periods for fault reset and pre-brief.

10. Teaching Points;

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference

TP1	1. Describe the operation of the CIWS Mount Servo Subsystem mode sequencing, to include: a. status go circuits: b. stow device operation; c. mount ready operation; d. mount rate gyro activate; and e. mount warning motion inhibit. 2. Mode control to include: a. Operational (mount assign) modes; b. Surface and LCP local control modes; c. Remote designate mode; d. Scan mode; e. Track mode; f. Local test modes: g. gyro test; h. gyro drift adjustment; and i. tach test. 3. Remote test modes to include: a. gyro loop test; b. rate test; c. current loop test; and	L P	10 6	

	d. lift/right power amplifier test.			

	e. Servo drive power on control; and f. Elevation/train servos: g. Servo input circuits; h. Power amplifier order circuits; i. Reference filters and function generators; 4. Voltage to firing angle pulse converters; 5. Power amplifiers; 6. Motor current sensor and control logic; and 7. Output circuits.			
TP2	1. Localize fault to LRU to include: a. symptom recognition; b. use of fault finding logic charts; c. appropriate test points; d. logical procedure; e. interpretation of schematics; f. Repair and align system; g. Tests and adjustments; h. Verify operational readiness.	P	12	

PO 003 Maintain Close In Weaspons System (CIWS)

EO 003.11 Maintain the CIWS Radar Subsystem

1. Performance. Maintain the CIWS Radar Subsystem.

2. Conditions.

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the CIWS Radar Subsystem, IAW refs, by:

   a. describing the CIWS Radar subsystem, to include:

   b. search receiving and signal processing subsystem;

   c. track receiving and signal processing subsystem;

   d. transmitter subsystem; and

   e. Radar Servo Interface (RSI) subsystem.

   f. conducting CIWS Radar subsystem functions, to include :

      (1) initializing;

      (2) localizing fault to LRU;

      (3) repairing and aligning system;

(4) verifying operational readiness; and

(5) powering down.

4. Time Required.

a. EO - 54 x 50 min periods.

5. Instructional Methods.

a. lecture; and

b. demonstration.

6. References.

a. A1 – A7, and

b. C1 – C20.

7. Training Aids.

a. CIWS System and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details.

a. EO 003.11 will be tested with PC 003 IAW Annex K.

9. Remarks.

a. Projects and exercises may be assigned;

b. MPT may be utilized; and

c. assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference

TP1	1. Describe the search receiving and signal processing subsystem, to include: a. search radar operation (sub modes). b. functional description to include: c. search antenna (3A1A2); d. associated waveguides; e. microwave receiver (3A1A9); f. signal processor; and g. search system mode sequencing. 2. functional circuit theory: a. search radar block diagram; b. WCC to TDP interface; c. TDP to subsystem interface; and d. subsystem to TDP interface;	L	10	
TP2	1. Describe the track receiving and signal processing subsystem, to include: a. major components: b. track antenna (3A1A7); c. microwave receiver (3A1A9); and d. Signal processor 2. functional description of track	L	14	

	subsystem to include: a. target handover; b. transition to track mode; c. target tracking; d. system interfaces: e. WCC to TDP interface; f. SP to subsystem interface; g. subsystem to SP interface; and h. SP to WCC interface. 3. track radar timing control; a. track microwave receiver; b. sum and delta IF and AGC; c. local oscillator control, doppler acquisition and tracking; d. range gate timing; e. position error signal demodulation; f. track and fire; g. CLAIMING A KILL; and h. moving target simulation.			

TP3	1. Describe the transmitter subsystem, to include: 2. A general description to include: a. purpose of the transmitter subsystem; b. controls and indicators; c. power and environmental requirements; and d. major components of the transmitter.	L	7	
	3. general circuit description to include: a. 115 VAC to the transmitter Mode Control Unit (TMCU); b. +28 VDC EM and +28 VDC ELX; c. 440 VAC ELX; and d. mode sequencing: e. battery off; f. standby; g. maintenance; h. air ready; i. search low PRF mode; j. search HPRF and track mode; k. alternate modes of operation; l. battleshort; and m. local control (RSMU);			

TP4	1. Describe the Radar Servo Interface (RSI) subsystem, to include: a. purpose; b. search/track servo capabilities; and c. major RSI components: d. search antenna assembly; e. scan drive assembly; f. search gimbal assembly;	L	7	

	g. track antenna assembly; h. track gimbal assembly; i. search/track microwave receiver; j. vertical reference unit; k. track RIGs; l. RSI assembly; and m. STSE assembly. 2. Description of the RSI Subsystem to include: a. track RIG heaters; b. RSI digital input; c. RSI digital output; d. track RIG activation circuits; e. search antenna scan drive; f. Vertical Reference Unit; g. search platform servo; h. track mode command logic; i. track RIGs; j. modes of operation: k. position mode; l. rate mode; and m. track mode. 3. track platform servo.			

TP5	1. Localize fault to LRU to include: a. symptom recognition; b. use of fault finding logic charts; c. appropriate test points; d. logical procedure; e. interpretation of schematics; f. Repair and align system; g. Tests and adjustments; h. Verify operational readiness	P	16 1 fault per subsystem	

PO 003 Maintain Close In Weaspons System (CIWS)

EO 003.12 Maintain the CIWS Surface Mode Subsystem

1. Performance. Maintain the CIWS Surface Mode Subsystem.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall maintain the CIWS Surface Mode Subsystem, IAW refs, by:

   a. describing the CIWS Surface Mode subsystem; and

   b. conducting CIWS functions, to include:

      (1) initializing;

      (2) localizing fault to LRU;

      (3) repairing and aligning system; (4) verifying operational readiness; and

      (5) powering down.

4. Time Required:

   a. EO - 44 x 50 min periods;

   b. EC - 1 X 50 min periods; and

c. PC - 17 X 50 min periods

5. Instructional Methods:

a. lecture;

b. Trainee practice

6. References:

a. A1 – A7, and

b. C1 – C20.

7. Training Aids:

a. CIWS System and/or MPT;

b. references;

c. audio visual aids; and

d. handouts.

8. Test Details:

a. EO 003.12 will be tested with EC 003.12 and PC 003 IAW Annex K;

b. EC 003.12 will be tested with a 1 period time limit; and

c. PC 003 and debrief to be conducted with a 90 min time limit. (Corrective measures start at minute 85).

9. Remarks:

a. MPT may be utilized,

b. assignments shall be given; and are to be completed during non-instructional time;

c. 1 period for review will be given prior to EC 003.12 ;

d. 24 periods of practical are allocated for overall CIWS system fault finding practice in preparation for the final PC;

e. system fault finding runs shall be conducted in pairs with lead changing after each fault; and

f. 2 periods for PC fault reset and prebrief.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference
TP1	1. Describe the operation of the CIWS PsuM Subsystem, to include: a. Acquisition Video Tracker (AVT): b. AVT CCA function; c. Video distribution CCA d. Convolver / frame store; e. Feature processor (FP) CCA. f. video tracker symbology: g. PsuM menus; and h. target acquisition.	L	2	
TP2	1. Describe the EOSS subsystem, to include: a. EOSS pedestal; b. EOSS electronics; c. system power amplifier (3A4A25); and d. SI/TCU (2A2) drawer assembly. 2. Describe the operating modes of the EOSS subsystem, to include : a. EO Environmental Control Mode; b. EO Standby mode; c. EO Ready mode; d. Fault isolation mode; e. Calibration mode; and f. EO Rate mode.			

	3. Describe the operation of the PTI subsystem, to include: a. PTI subsystem overview; b. PTI subsystem functions c. infrared light ray acquisition and focus; d. telescope wash and wipe; e. vertical (field) mirror scanning; f. horizontal (line) mirror scanning; g. infrared signal detection and preamplification; h. video signal channel processing; i. digital electronics; j. serial fibre optic transmission; and k. Built-in Test (BIT).	L	3	
TP3	1. Installation/removal/packing of the EOSS and PTI subsystems.	L P	2 5	

TP4	1. Localize fault to LRU to include: a. symptom recognition; b. use of fault finding logic charts; c. appropriate test points; d. logical procedure; e. interpretation of schematics; f. Repair and align system;	P	32 2 faults for PsuM subsystem, 2 faults for system Trainee practice	
	g. Tests and adjustments; and h. Verify operational readiness			

PO 003 Maintain Close In Weaspons System (CIWS)

EO 003.13 Perform Pre/Pose Firing Check on the CIWS

1. Performance. Perform Pre/Post Firing Check on the CIWS.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS; and

      (7) assistance.

   b. Denied:

      (1) Nil.

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The Trainee shall perform pre / post firing routines on the CIWS, IAW refs, by:

   a. explaining INO requirements;

   b. performing Pre firing routine; and

   c. performing Post firing routine.

4. Time Required:

   a. EO = 14 x 50 min periods.

5. Instructional Methods:

   a. lecture; and

   b. Trainee practice.

6. References:

   a. A1 – A7, and

   b. C1 – C20.

7. Training Aids:

   a. CIWS System and/or MPT;

   b. references;

   c. audio visual aids; and

   d. handouts.

8. Test Detail:

   a. N/A.

9. Remarks:

   a. N/A.

10. Teaching Points:

   a. L=Instructor Lecture, D=Instructor Demo, P=Trainee Practice

	Description	Method	Time	Reference
TP1	1. Explain INO requirements	L	1	
TP2	1. Perform pre firing routine	P	6.5	

TP3	1. Perform post firing routine	P	6.5	

PO 004 – Operate the Close In Weapons System (CIWS)

EO 004.01 Operate the CIWS

1. Performance. Operate the CIWS.

2. Conditions:

   a. Given:

      (1) supervision;

      (2) references;

      (3) DWAN;

      (4) tools and test equipment;

      (5) PPE;

      (6) CIWS and/or NCOT; and

      (7) assistance.

   b. Denied:

      (1) Nil

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night

3. Standard. The Trainee shall operate the CIWS, IAW refs, by:

   a. applying policy order from Stand by Mode;

   b. conducting PSOT 14;

   c. conducting PAC drills and procedures;

   d. conducting CIWS surface engagements; and

   e. conducting CIWS are engagements

4. Time Required:

   a. EO - 20 x 50 min periods; and

   b. PC – 8 x 50 min periods;

5. Instructional Methods:

   a. Lecture;

   b. Demonstration;

   c. Trainee Practice.

6. References:

   a. A1 – A7, and

   b. C1 – C20.

7. Training Aids:

   a. CIWS and/or NCOT;

   b. audio visual aids;

   c. references; and

   d. handouts.

8. Test Details:

   a. EO 004.01 will be tested with PC 004, IAW Annex K; and

   b. PC 004 and debrief will be conducted with a 1 period time limit. (corrective measures starting at minute 45).

9. Remarks:

   a. Projects and exercises may be assigned;

   b. MPT may be utilized; and

c. assignments are to be completed during non-instructional time.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	1. Perform close up and clear IAW policy order, IAW refs, by: a. Apply capabilities and limitations in accordance with engagement criteria; b. From LCP to include; (1) Insert 'No engage sectors;' (2) Insert 'Max/Min Range rates; and' (3) Insert First Hit parameters. c. Describe Controller functions; d. From Controller to include: (1) Insert 'No engage sectors;' (2) Insert 'Max/Min Range rates; and' (3) Insert First Hit parameters.	D P	0.5 2	
TP2	1. Conduct PSOT 14	D P	0.5 2	

TP3	1. Conduct PAC drills and procedures	D P	1 4	
TP4	1. Conduct CIWS Air Engagements	D P	1 4	
TP5	1. Conduct CIWS Surface Engagements	D P	1 4	

PO 005 – Maintain Combat Management System

EO 005.01 Describe the Combat Management System (CMS)

1. Performance. Describe the CMS.

2. Conditions:

   a. Given:

      (1) CCMS and/or MPT;

      (2) tools and test equipment;

      (3) replacement parts;

      (4) references;

      (5) software;

      (6) supervision; and

      (7) assistance as required.

   b. Denied:

      (1) Nil

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The trainee shall describe the CMS, IAW refs, by:

   a. system components;

   b. capabilities and limitations ;

   c. theory of operation;

d. initial power distribution; and

e. indicators and controls.

4. Time:

a. EO = 10 x 50 min periods.

5. Method / Media:

a. lecture.

6. References:

a. A1 – A7; and

b. D1 – D4.

7. Training Aids:

a. a CMS and/or MPT;

b. audio visual aids;

c. references; and

d. handouts.

8. Test Details:

a. EO 005.01 will be tested as part of PC 005 IAW Annex K.

9. Remarks:

a. For the purpose of this EO, the student is assumed to have a working knowledge of CMS operation, initialization and maintenance. This training should be focussed on the higher functions of CMS.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference

TP1	1. Describe system components to include; a. mission critical enclosures; b. multi-function workstation; and, c. local area network	L	2	
TP2	1. Capabilities and limitations to include; a. Cabinet enclosures; b. remote access server; c. cabinet cooling system; and d. MCE cabinets	L	2	
TP3	1. Theory of operation to include; a. layered system of sensors; b. information exchange systems(IES); c. tracking tools; and d. decision aids and weapons to include: (1) detect (2) locate; (3) track; (4) identify; (5) assess; and (6) engage threats.	L	2	

TP4	1. Power Distribution to include; a. Breaker and switchboard connection; b. MCE power; c. MFW power; and d. subassembly voltage input	L	2	
TP5	1. Indicators and controls to include; a. Common MCE Components; b. MCE 1 and 2 Subassemblies; c. MCE 3 and 4 Subassemblies; d. MCE 5 Subassemblies; e. MCE 6 Subassemblies; f. MCE 7 Subassemblies; g. MCE 8 Subassemblies; h. MCE 11 Subassemblies; and i. MCE 12 Subassemblies	L	2	

PO 005 – Maintain Combat Management System

EO 005.02 Initialize the Combat Management System (CMS)

1. Performance. Initialize the CMS.

2. Conditions:

   a. Given:

      (1) CMS and/or MPT;

      (2) tools and test equipment;

      (3) replacement parts;

      (4) references;

      (5) software;

      (6) supervision; and

      (7) assistance as required.

   b. Denied:

      (1) Nil

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The trainee shall initialize the CMS, IAW refs, by:

   a. powering up;

   b. conducting equipment functional checks;

c. loading CMS software;

d. verify indicator status;

e. report equipment status to supervisor; and

f. powering down.

4. Time:

a. EO = 10 x 50 min periods.

5. Method / Media:

a. lecture

b. demonstration; and

c. PC.

6. References:

a. A1 – A7; and

b. D1 – D4.

7. Training Aids:

a. a Combat Management System and/or MPT;

b. audio visual aids;

c. references; and

d. handouts.

8. Test Details:

a. EO 005.02 will be tested as part of PC 005 IAW Annex K.

9. Remarks:

a. practice Flash up and faultfinding will be conducted in teams of 2; and

b. EC/PC checks will be carried out individually.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference

TP1	1. Initialize CMS to include; a. Preliminary checks on associated equipment; b. Power up; c. Checks during and after power on; d. Load and verify software; e. Conduct system functional checks; f. Verify operational readiness; g. Report System Status to supervisor; and h. power down	D P	2 8	

PO 005 – Maintain Combat Management System

EO 005.03 Maintain the Combat Management System (CMS)

1. Performance. Maintain the CMS.

2. Conditions:

   a. Given:

      (1) CMS and/or MPT;

      (2) tools and test equipment;

      (3) replacement parts;

      (4) references;

      (5) software;

      (6) supervision; and

      (7) assistance as required.

   b. Denied:

      (1) Nil

   c. Environmental:

      (1) any shipboard conditions at sea or alongside; and

      (2) day or night.

3. Standard. The trainee shall maintain the CMS, IAW refs, by:

a. powering up;

b. localize the fault to the LRU;

c. repair;

d. verify operational readiness;

e. report equipment status to supervisor; and

f. powering down.

4. Time:

a. EO = 10 x 50 min periods; and

b. PC = 5 X 50 min periods.

5. Method / Media:

a. Lecture

b. Demo; and

c. PC.

6. References:

a. A1 – A7; and

b. D1 – D4.

7. Training Aids;

a. Combat Management System and/or MPT;

b. audio visual aids;

c. references; and

d. handouts.

8. Test Details:

a. EO 005.03 will be tested with PC 005 IAW Annex K.

9. Remarks:

a. Practice Flash up and faultfinding will be conducted in teams of 2;

b. EC/PC checks will be carried out individually.

10. Teaching Points:

a. L=Instructor Lecture, D=Instructor Demo, P=Student Practice

	Description	Method	Time	Reference
TP1	1. Maintain CMS, by; a. symptom recognition; b. localize fault to LRU c. use of fault finding logic charts; d. logical	P	10	
	procedure; e. interpretation of schematics; f. verify operational readiness; g. tests and alignments; h. describe removal and replacement of LRU's; and, i. verify operational readiness			

QUALIFICATION STANDARD ANNEXES

MARPAC RDIMS #1424172 v2
Chap 1 & 2 approved 09/09/ 2024
Chap 3 & 4 approved 09/09/ 2024

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A1		Class Specific Safety and Environment Management System (SEMS)
A2	B-GN-007-RCN/FP-002 CFCD 133	RCN ERT Manual
A3	C-01-000-000/DA-001	NDID Coding System
A4	SSOs	Class Specific Ship's Standing Orders
A5	CF H Svcs Gp Instruction 4440-01	Chemical Hazards Surveillance Program
A6		Class Specific ERT Pack
A7	DAOD 4003-0	Environmental Protection and Stewardship
B		**Bofors 57mm Main Gun System Publications**
B1	C-70-270-002/DF-001	57mm mk3 Ammunition and Range Tables (PFHE)
B2	C-70-270-002/DF-002	57mm mk3 Ammunition and Range Tables (HCER)
B3	C-70-270-002/DF-003	57mm mk3 Ammunition and Range Tables (3P)
B4	C-70-270-002/NH-001	57mm Mk3 Error Codes
B5	C-70-270-002/MD-002	57mm Mk3 Functional Description
B6	C-70-270-002/MP-001	57mm Mk3 Maintenance Instructions
B7	C-70-270-002/MB-001	57mm Mk3 Operators Manual
B8		57mm Mk3 Technical Drawings Electrical and Hydraulic
B9	C-70-270-002/MD-001	57mm Mk3 Technical Description
B10	C-70-270-002/MY-001	57mm Mk3 Illustrated Parts Breakdown
B11	C-70-270-002/NY-001	NAVAL PREVENTIVE MAINTENANCE SCHEDULE (WEAPONS) MAIN GUN SYSTEM 57 MM MK III
B12	C-70-270-002/NY-Z01	NAVAL PREVENTIVE MAINTENANCE SCHEDULE (Weapons Engineering Armament) MAIN GUN SYSTEM 57 MM MK III
C		A **Close-In Weapon System Publications**
C1	C-70-374-000/ME-001	Close-In Weapon System (CIWS) MK 15 MOD 11, 12, 13, 14, 21, 22, AND 25 PHALANX BLOCK 1B Light-Off And Checkout Procedures
C2	C-70-374-000/MS-003	Close-In Weapon System MK 15 MODS 21, 22, and 25 BLOCK 1B Handbook
C3	C-70-374-000/MS-004	Comprehensive Maintenance Manual Fiber Optic Cable Termination, Assembly, And Test Procedures For BLOCK 1B PHALANX Close-In Weapon System (CIWS) MK15 MOD 21,22, AND 25

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
C4	C-70-374-000/MS-006	Maintenance Peculiar Support Equipment For CIWS MK 15 (PHALANX)
C5	C-70-374-000/MS-007	Intermediate Maintenance Manual M61A1 20mm (CIWS) Gun And Ammo Handling System
C6	C-70-374-000/NY-001	Naval Preventive Maintenance Schedule (Weapons) Close-In Weapon System MK 15 MODS 21 and 22 BLOCK 1B
C7	C-70-374-000/NY-Z01	Naval Preventive Maintenance Schedule (Weapons) Close-In Weapon System MK 15 MOD 21 and 22 BLOCK 1B
C8	C-70-374-000/RT-001	Readiness Trial For Close In Weapon System (CIWS) MK 15 MOD 21-22 BLOCK 1B - PART 1: Physical Inspection
C9	C-70-374-000/RT-002	Readiness Trial For Close In Weapon System (CIWS) MK 15 MOD 21-22 BLOCK 1B Functional Test
C10	C-70-374-000/RT-004	Readiness Trial For Close In Weapon System (CIWS) MK15 MOD 21-22 BLOCK 1B
C11	C-70-374-000/TB-	002 Technical Bulletin (Navy) Phalanx Close-In Weapon System MK 15 MODS 21 and 22 BLOCK 1B 20 Millimetre Ammunition Types Adjustment
C12	C-70-374-000/VB-001	Material And Equipment Protection Instructions For Close In-Weapon System (CIWS) MK 15 BLOCK 1A and BLOCK 1B
C13	C-70-374-005/DL-001	TECHNICAL MANUAL - REFERENCE DIAGRAMS (FUNCTIONAL) - PHALANX CLOSE IN WEAPON SYSTEM MK 15 MOD 21-22 BLOCK 1B BASELINE 2 APPLICABLE TO HFX CLASS, NFS(A) AND NFS(P)
C14	C-70-374-005/DM-001	TECHNICAL MANUAL ILLUSTRATED PARTS BREAKDOWN BOOKS 1 TO 4 OF 4 PHALANX CLOSE-IN WEAPON SYSTEM MK 15 MODS 21-22 BLOCK 1B BASELINE 2 APPLICABLE TO HFX CLASS, NFS(A) AND NFS(P)
C15	C-70-374-005/ME-001	TECHNICAL MANUAL INTERMEDIATE MAINTENANCE - PHALANX CLOSE IN WEAPON SYSTEM MK 15 MOD 21-22 B/L 2 LIGHT-OFF AND CHECKOUT PROCEDURES APPLICABLE TO HFX CLASS, NFS(A) AND NFS(P)
C16	C-70-374-005/MM-001	TECHNICAL MANUAL - PHALANX CLOSE IN WEAPON MK 15 MOD 21-22 BLOCK 1B BASELINE 2 MAINTENANCE REQUIREMENTS CARDS (MRC) AND MAINT INDEX PAGE (MIP) APPLICABLE TO HFX CLASS, PRO CLASSES, NFS(A) AND NFS(P)

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
C17	C-70-374-005/MS-003	TECHNICAL MANUAL HANDBOOK - CLOSE-IN WEAPON SYSTEM MK 15 MODS 21, 22, AND 25 BLOCK 1B BASELINE 2 APPLICABLE TO HFX CLASS, NFS(A) AND NFS(P)
C18	C-70-374-005/RT-001	READINESS TRIAL FOR PHALANX CLOSE-IN WEAPON SYSTEM (CIWS) MK 15 MOD 21-22 BLOCK 1B BASELINE 2 PART 1: PHYSICAL INSPECTION APPLICABLE TO HFX, PRO CLASSES AND NFS
C19	C-70-374-005/RT-002	READINESS TRIAL FOR PHALANX CLOSE-IN WEAPON SYSTEM MK 15 MOD 21-22 BLOCK 1B BASELINE 2 PART 2 FUNCTIONAL TEST APPLICABLE TO HALIFAX CLASS
C20	C-70-374-005/RT-004	READINESS TRIAL FOR PHALANX CLOSE-IN WEAPON SYSTEM (CIWS) MK 15 MOD 21-22 BLOCK 1B BASELINE 2 PRE-ACTION CALIBRATION AND SURFACE FIRINGS TEST APPLICABLE TO HALIFAX CLASS
D		**Combat Management System Publications**
D1	C-60-236-000/MD-001	Equipment Description
D2	C-60-236-000/MM-001	First Line Maintenance
D3	C-60-236-000/NK-001	PM Performance Tests
D4	C-60-236-000/NY-001	PM Schedule
E		**Combat Management System Publications**
E1	C-58-647-000/DW-001	Drawings and Schematics
E2	C-58-647-000/MB-001	Operating Instructions
E3	C-58-647-000/MD-001	System Description
E4	C-58-647-000/MD-002	Functional Description
E5	C-58-647-000/MN-001	Maintenance Instructions
E6	C-58-647-000/MY-001	Illustrated Parts Breakdown
E7	C-58-647-000/NY-001	Scheduled Maintenance
E8	C-58-647-000/NY-Z01	Scheduled Maintenance
E9	C-58-647-000/RT-001	Physical Readiness Trial
E10	C-58-647-000/RT-002	Functional Readiness Trial

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

Task #	Task	PO
Duty Area E	**MASS**	
E001	Initialize a Rocket Launching System to an operational state. (e.g. Shield, NULKA, SRBOC)	
E002	Load/unload a Rocket Launching System	
E003	Monitor a Rocket Launching System performance	
E004	Operate a Rocket Launching System	
E005	Perform a Rocket Launching System functional check	
E006	Perform a Rocket Launching System PM activity (including a test/trial/inspection)	

E007	Perform a Rocket Launching System post-firing check/routine	
E008	Perform a Rocket Launching System pre-firing check/routine	
E009	Perform post-repair calibration /testing /tuning/ alignment of a Rocket Launching System	
E010	Remove unexploded/damaged ordnance from a Rocket Launching System	
E011	Repair/refurbish/overhaul a faulty component from a Rocket Launching System	
E012	Calibrate/test/tune/align a newly installed Rocket Launching System component	
E013	Conduct preliminary fault diagnostic on a Rocket Launching System (Tech of the Watch)	
E014	Configure a Rocket Launching System to meet an operational requirement	
Duty Area	**57 mm Gun**	

C		
C001	Operate a Naval Gun System	
C002	Remove a faulty component from a Naval Gun System	
C003	Diagnose a Naval Gun System fault to the Lowest Repairable Unit	
C004	Configure a Naval Gun System to meet an operational requirement	
C005	Carry out a naval gun misfire procedure	
C006	Calibrate/test/tune/align a newly installed Naval Gun System component	
C007	Remove unexploded/damaged ordnance from a Naval Gun System	

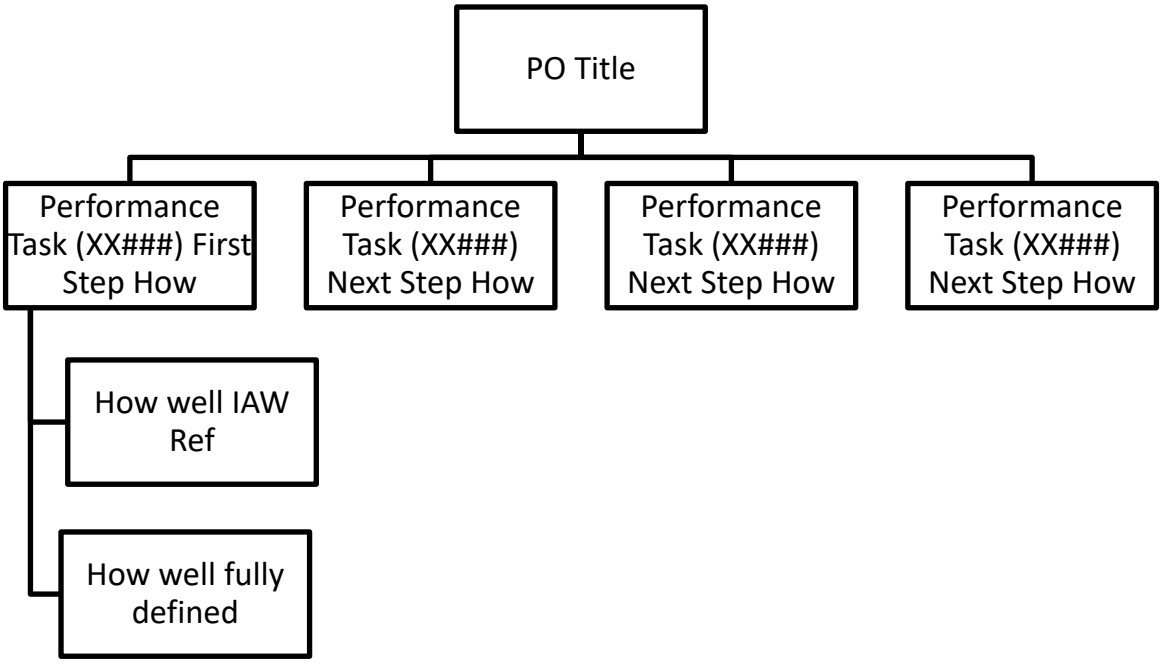
C008	Perform the duties of the Captain of the Turret (COT)	
C009	Perform post-repair calibration/testing/tuning/alignment of a Naval Gun System	
C010	Perform a Naval Gun System PM activity (including a test/trial/inspection)	
C011	Perform a naval gun post-firing check/routine	
C012	Perform a naval gun pre-firing check/routine	
C013	Perform a Naval Gun System functional check	
C014	Monitor a Naval Gun System performance	
C015	Load/unload a naval gun	
C016	Install a new or repaired/refurbished/overhauled Naval Gun System component	
C017	Initialize a Naval Gun System to an operational state	
C018	Conduct preliminary fault diagnostic on a Naval Gun System (Tech of the Watch)	
Duty Area C	**CIWS**	
C019	Operate a CIWS	

C020	Initialize a Close In Weapon System (CIWS) to an operational state	
C021	Monitor a CIWS performance	
C022	Perform a CIWS functional check	
C023	Load/unload a CIWS	
C024	Perform a CIWS pre-firing check/routine	
C025	Perform a CIWS post-firing check/routine	
C026	Diagnose a CIWS fault to the Lowest Repairable Unit	

C027	Conduct preliminary fault diagnostic on a CIWS (Tech of the Watch)	
C028	Perform post-repair calibration/testing/tuning/alignment of a CIWS	
C029	Calibrate/test/tune/align a newly installed CIWS component	
C030	Remove unexploded/damaged ordnance from a CIWS	
C031	Perform the duties of the Remote Control Panel Operator	
C032	Perform duties of the Local Control Panel Operator (LCPO)	
C033	Perform a CIWS PM activity (including a test/trial/inspection)	
Duty Area C	**Operate CIWS**	
C034	Operate a CIWS	
C035	Configure a CIWS to meet an operational requirement	
C036	Carry out a CIWS misfire procedure	
C037	Perform a CIWS functional check	
C038	Perform the duties of the Remote Control Panel Operator	
C039	Perform duties of the Local Control Panel Operator (LCPO)	
Duty Area O	**CMS**	
O001	Initialize the Combat Management System (CMS) to an operational state	
O002	Monitor the CMS performance	
O003	Install a new or repaired/refurbished/overhauled CMS component	

O004	Operate the CMS	
O005	Perform the CMS functional checks	
O006	Perform post-repair calibration/testing /tuning/alignment of the CMS	
O007	Calibrate/test/tune/align a newly installed CMS component	
O008	Configure a CMS to meet an operational requirement	
O009	Diagnose a CMS fault to the Lowest Repairable Unit	
	DRMIS	
	As per DRMIS QSP	

ANNEX C – QS SCALAR New requirement under new format.



ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Dates: 12 -21 Oct 10

Location:

CFB Esquimalt Chief's and PO's Mess, Pacific Station Board Members:

SURNAME	INITIALS	MOSID	UNIT
STONIER	P.C.	00119	D MAR PERS 3-2-3
MAGEE	M.G.	00119	DMTE
PARSONS	J.D.	00017	MARPACHQ
COBEAN	S.A.	00017	CFNES
WALKER	R.W.	00017	CFFSE
LEMIEUX	D.P.	00017	CFFSE
POIRIERPOTVIN	J.R.P.L.	00017	HMCS REGINA
JONES	B.D.	00299	CFFSE
GIROUX	C.J.	00211	CFFSE

Board Objective:

To create the Job Qualification for the W Eng Armament System maintainer.
Specification Task – No Train Decisions:

The tasks selected for training are detailed at Annex B. Tasks not selected for training are listed below with substantiation:

TASK SERIAL	TASK ELEMENT	SUBSTANTIATION
TASK SERIAL	**TASK ELEMENT**	**SUBSTANTIATION**

Discussions / Decisions:

DAY 1 – 12 OCT 10

1. The Chairman, CPO1 J.D. Parsons, commenced the board by providing the mandate, methodology, and a description of a QSP and its component parts. He discussed the Mandate, Board Agenda, Schedule, Roles & Responsibilities, Board Conduct, Record of Proceedings, and CFITES. The Mandate is to "create the QSP for the W Eng Tech Armament Maintainer Training' that results in qualified personnel capable of carrying out duties commensurate with their rank and MOSID requirements." DMTE CPO1 M. Magee briefed the board on decisions and procedures used by previous W Eng Tech maintainer QSP boards and demonstrated the template to be used. DMARPERS, CPO1 P. Stonier, discussed the W Eng Tech training philosophy including "Conceptual knowledge", vs. "Detailed knowledge", the trainee responsibility to be an active learner outside of the training establishment, and equipment training aimed at system familiarization as opposed to producing day one subject matter experts.

2. CPO1 Magee stated that DMTE requires a finished product from this board. By 15 Dec 2010, need to produce a trainable QSP. 1 Apr 2011 is the start up date for the first serial for the Jr W Eng Tech Module 1.

3. CPO1 Magee discussed scalars, which take all tasks and puts them in a logical sequence for training. He also discussed the new trade progression system. CPO1 Stonier emphasized that the training must address a specific job requirement, as opposed to training everyone for every job they may hold throughout their career in the trade. Where specific or unique skills are required for a limited number of jobs these will be addressed on a case by case basis (Speciality Specification).

4. The board reviewed Chapter 1 and completed Chapter 2 Performance Objectives then adjourned for lunch.

5. During the afternoon the board completed three Enabling Objectives (EO) for the first Performance Objective (PO), Shield Off board ECM Maintenance.

DAY 2 – 13 Oct 10

6. The board reviewed the ROP and the work of the previous day.

7. The remainder of the morning was spent on completing the first PO and starting the second PO 57mm Gun Maintenance.

8. There was some discussion about a trainee project for EO 002.005, 57mm Gun Loading System.

DAY 3 – 14 APR 10

9. Reviewed ROP and previous days work.

10. Reviewed Chapter 4.

DAY 4 – 15 APR 10

11. Reviewed ROP.

12. Developed ECs and PC for 57mm Gun.

13. The board recommended that pre/post firing procedures be added to the Fire Control Maintainer and W Eng Tech Leading Seaman RQ QSPs for missiles and torpedoes. This was actioned by DMTE. These Occasional Maintenance Routines are essential weapons safety procedures. There is no guarantee that the trainee will have received this training prior to these courses.

14. Completed review of 57mm Gun PO.

15. The board discussed the decreased length in training vs. the ship's expectation of the member. The board was reminded of the training philosophy by DMTE and DMARPERS "the idea is to teach the students to think on their own, vice teach each piece of equipment in detail".

DAY 5 – 18 Oct 10

16. Reviewed ROP.

17. The board started working on the CIWS PO.

DAY 6 – 19 Oct 10

18. Reviewed ROP.

19. The board completed the CIWS Maintenance and the CIWS Operator POs.

20. Annex G was completed.

21. The board started review of the entire document for formatting.

DAY 7 – 20 APR 10

22. Reviewed ROP.

23. The board completed the review and correction of the entire document for formatting.

24. The board completed the reference and acronym lists.

DAY 8 – 21 Oct 10

25. Reviewed ROP.

26. The board reviewed the entire document for content and timing.

27. The board presented the completed QSP to DMTE. The Chairman thanked all the members for their expertise and dedicated efforts and closed the board.

19.
Lessons Learned:

28. When using the QSP Wiki, it is imperative that all SME's use the exact same format. This will eliminate multiple corrections on final document review.

29. The Wiki is incapable of producing tables that are easily edited. Therefore, you must cut and paste content into the final document. It was found that the Wiki's primary strength is for producing content concurrently, as opposed to formatting.

30. The board noticed that it was sometimes necessary to double-save the Wiki in order to save documents.

31. All supporting documentation, i.e. Master Lesson Plans, timing information, and reference lists should be available to the board.

Guidance to Instructional Designers:

32. Ideally, all practice fault-finding is conducted in pairs. To conduct one practice fault the board allotted 4 hours for a course of 8 based on a 45 minute time limit. If 20 periods are allotted for practice faults, the board's intention is to have 5 practice faults for that period. If time allows, the instructors are free to insert additional faults, or consider individual faults. Instructors are encouraged to assign in class projects for trainees waiting for hands on equipment practice time.

33. Instructors are encouraged to give homework in the form of reading assignments. These may be validated by a short quiz or question and answer sessions.

34. The level of detail taught on any specific Teaching Point must be commensurate with the time allotted.

35. The required trainee project for Enabling Objective 002.05 shall consist of a hand written detailed description of step-by-step mechanics of cycling a round from the main magazine to the case ejection chute.

Follow-up Action:

36. On completion, DMTE will distribute the completed QSP to board members, TDO, and supervisors for approval.

Other Considerations (if applicable)

TRAINING PLAN ANNEXES

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

2. Facilities/Trainers

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

3. Ammunition

SER	LOG GUIDE	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

4. Weapons

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

5. Specialized/Technical Equipment

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

6. Transport/Vehicles

SER		NSN	DESCRIPTION		QTY	REMARKS
a.		XXX-XX-XXX-XXXX				
b.		XXX-XX-XXX-XXXX				
c.		XXX-XX-XXX-XXXX				
	NOTES:					

7. Communication Equipment/Frequencies

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

8. Training Aids

SER	DESCRIPTION		QTY	REMARKS
a.				

b.				
c.				
NOTES:				

9. POL Products

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

10. General Purpose Stores

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

11. Stationery

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

12. Personal Equipment

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

ANNEX F – COSTING INFORMATION New requirement under new format values in table not accurate for Arm example only.

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: EXAMPLE		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	14	$1,000.0		$14,000	$14,000	$70,000.00
R&Q	14	$36.0	20	$10,080	$10,080	$50,400.00
Incidentals	14	$17.5	20	$4,900	$4,900	$24,500.00
Subtotal				**$28,980**	**$28,980**	**$144,900.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	16	$50.0	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.00	15	…	…	…
Computer requirements	16	$1,000.0	1	$16,000	$0	$16,000.00
IT Equip (Projector, Smart board etc.)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.0	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc.)	15	$30.0	1	$450	$450	$2,250.00
Subtotal				**$28,265**	**$1,265**	**$33,325.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			

Instructor (PO1/WO) (incremental)2	2	$223.0	20	$8,920	$8,920	$44,600.00
Incremental Civilian Instructor		XXX				
		subtotal		**$8,920**	**$8,920**	**$44,600.00**
	Total Estimated Costs			**$66,165**	**$39,165**	**$222,825**

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module.

3. The table below is available to use as a template for calculations.

Allocation of ATT	**Minutes**
Actual Training Time	
ATT Sub-total	**640.4x50min**
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	
Training Administrative Time (TAT)	
Training Travel Time (TTT)	
Physical Fitness (PT)	
Sub-total	
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**106**

OPTIMUM TRAINING SCHEDULE *New requirement under new format.*

Week 1 Time	Day 1	Day 2	Day 3	Day4	Day 5
Period 1 **0800-0850**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 2 **0900-0950**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 3 **1000-1050**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 4 **1100-1150**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 5 **1300-1350**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 6 **1400-1450**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 7	**EO/EC /PC/Admin** Title(s) / Activity				

Week 1 Time	Day 1	Day 2	Day 3	Day4	Day 5
1500-1550					
Period 8 **1600-1650**	**EO/EC /PC/Admin** Title(s) / Activity				

ANNEX H – ASSESSMENT PLAN

EC/PC	Scope Covered	Purpose	How	When	Resources	Pass Mark	Impact
EC 001.03	EO 001.03	To evaluate the Trainees knowledge to maintain the MASS__	Written test	O/C of EO 001.03	MASS Publications & Stationary	60%	Re-attempt TRB
PC 001	EO 001.01 EO 001.02, and EO 001.03	To evaluate the Trainees ability to maintain the MASS	Practical test	O/C of 001.03	MASS System	60%	Re-attempt TRB
EC 002.01	EO 001.01	To evaluate the Trainees knowledge to describe the 57mm gun	Written	O/C of EO 001.02	57mm publications & Stationary	60%	Re-attempt TRB
EC 002.03	EO 001.03	To evaluate the Trainees knowledge on the operation of the 57mm gun	Written	O/C of EO 002.03	57mm publications & Stationary	60%	Re-attempt TRB
EC 002.04	EO 002.04	To evaluate the Trainees knowledge on the operation of the 57mm gun	Written	O/C EO 002.04	57mm publications & Stationary	60%	Re-attempt TRB

EC 002.05	EC 002.05	To evaluate the Trainees knowledge on the operation of the 57mm gun	Written	O/C EO 002.05	57mm publications & Stationary	60%	Re-attempt TRB
EC 002.06	EC 002.06	To evaluate the Trainees knowledge on the operation of the 57mm gun	Written	O/C EO 002.06	57mm publications & Stationary	60%	Re-attempt TRB
EC 002.07	EC 002.07	To evaluate the Trainees knowledge on the operation of the 57mm gun	Written	O/C EO 002.07	57mm publications & Stationary	60%	Re-attempt TRB
EC 002.08	EC 002.08	To evaluate the Trainees knowledge on the operation of the 57mm gun	Written	O/C EO 002.08	57mm publications & Stationary	60%	Re-attempt TRB
EC 002.09	EC 002.09	To evaluate the Trainees knowledge on the operation of the 57mm gun	Written	O/C EO 002.09	57mm publications & Stationary	60%	Re-attempt TRB
PC 002	EO 002.01 thru EO 002.09	To evaluate the Trainees ability to maintain the 57mm gun	Practical	O/C EO 002.09	57mm System	60%	Re-attempt TRB

EC 003.10	EO 003.01 Thru EO 003.10	To evaluate the Trainees knowledge on the operation of the CIWS	Practical	O/C EO 003.10	CIWS publications & Stationary	60%	Re-attempt TRB

EC 003.12	EO 003.11 Thru EO 003.12	To evaluate the Trainees knowledge on the operation of the CIWS	Written	O/C EO 003.12	CIWS publications & Stationary	60%	Re-attempt TRB
PC 003	EO 003.01 Thru EO 003.12	To evaluate the Trainees ability to maintain the CIWS	Practical	O/C EO 003.12	CIWS System	60%	Re-attempt TRB
PC 004	PO 004	To evaluate the Trainees ability to operate the CIWS	Practical	O/C of EO 004.02	CIWS System	60%	Re-attempt TRB
PC 005	PO 005	To evaluate the Trainees ability to maintain the CMS	Practical	O/C of EO 005.03	CMS	60%	Re-attempt TRB
PC 006		AS DRMIS QSP					

ANNEX I – ASSESSMENT CHECKLISTS From old QSP

PART I - ADMINISTRATIVE INSTRUCTIONS

1. This is a generic list for administering a PC/EC.

2. The Assessor shall:

   a. ensure all publications, test equipment and equipment are serviceable or Trainee informed of deficiencies;

   b. ensure the Trainee s observe all safety precautions;

   c. test Trainee s individually;

   d. supervise and provide physical assistance. as required;

   e. provide a pre and post assessment waiting area; and

   f. review Annex K with the Trainee .

3. If, after allotted time, the Trainee is not showing sufficient progress the Assessor may allot 15 additional minutes to complete the PC/EC with the loss of two points. For each additional 5 minutes after the initial 15 minutes will result in two points being lost. The maximum extra time granted will be 30 minutes. If the student does not find the fault after the additional 30 minutes, the instructor will stop the PC/EC and calculate the final mark.

4. The Assessor is to ensure that the best use of time allotted for PCs/ECs is made by facilitating the PC/EC process. For example:

5. if the Trainee is about to perform a logical but time consuming procedure the Assessor should supply the results and have the Trainee resume without penalty to the student however, if the action is not justified by the Trainee it is considered incorrect procedure; or

6. after a maximum of 10 minutes of inactivity the Assessor shall provide assistance..

7. Pass mark is 60%.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. You are given the following:

   a. References;

   b. Course notes;

   c. Job sheets;

   d. Work sheets;

e. Projects; and

f. Equipment and/or MPT.

2. IAW the references and PC/EC list, you shall:

a. Power up;

b. Conduct system functional checks;

c. Load software or crypto as required;

d. Localize fault to LRU;

e. Repair;

f. Verify operational readiness;

g. Report status to assessor; and

h. Power down.

3. Prior to performing a test or measurement you shall inform the Assessor of your intention.

4. IAW the references and PC/EC List, you shall complete the PC/EC:

a. Within allotted time IAW Chapter 4;

b. Individually;

c. With supervision; and

d. Physical assistance. as required.

5. IAW the references, you shall observe all safety precautions. A safety infraction or violation will result in instructor intervention and the instructor may terminate PC/EC based on the severity of the violation.

6. Pass mark is 60%.

7. Do you have any questions?

Trainee Sign: _____Date:_____

Instructor Sign:_____ Date:_____

PC/EC EVALUATION SHEET			
Equipment:		**Name:**	
PO Number:	**Start:**	**Stop:**	**Total:**
Course:			
Note: When allocating a score use only whole numbers. Circle the appropriate number in score column			

Initialization (10%)		
Score	**Justification**	**Remarks**
1 - 2	Did not follow correct flash up procedure. Flash up faults not recognized	
3 - 4	Followed procedure but flash up faults not recognized	
5 - 6	Followed flash up procedure. After some hesitation, corrected flash up faults	
7 - 8	Followed Flash Up procedure. Recognized and corrected flash up faults	
9 - 10	Confidently Flashed Up, efficiently corrected flash up faults.	

Use of Test Equipment (10%)		
Score	**Justification**	**Remarks**
1 - 2	Did Not select appropriate TE for Fault/ Did not demonstrate correct TE operating procedures	
3 - 4	Hesitant with selection of TE/Unsure of TE set up	

5 - 6	Selected correct TE/ Demonstrate weak operating procedures	
7 - 8	Select Correct TE/Properly utilized TE	

9 - 10	Selected Correct TE considering best options/Demonstrated thorough knowledge of operation	
Fault Finding Procedures (30%)		
Score	**Justification**	**Remarks**
1 - 2	Hesitant to begin Fault Finding./Did not recognize fault indications	
3 - 4	Did not correctly interpret Fault Indications; BITE; Test Equipment Readings; Took the wrong path and did not recover.	
5 - 6	Minor inconsistencies in interpreting fault indications. May have taken wrong path but recovered. Employed logical faultfinding.	
7 - 8	Recognized and Correctly interpreted fault indications. Employed Logical Faultfinding.	

9 - 10	Efficiently interpreted Fault Indications and evaluated all symptoms. Found fault with minimum number of steps.	
System Knowledge (30%)		
Score	**Justification**	**Remarks**
1 - 2	System knowledge did not support faultfinding. Could not answer questions from instructor. Could not properly use supporting documents	
3 - 4	System knowledge weak. Could not answer all questions. Struggled with documentation to support faultfinding. Could not fully explain fault	
5 - 6	System knowledge weak in some areas. Answered most questions from instructor and gave general explanations regarding the fault.	
7 - 8	Good system knowledge. Used of supporting documentation aided in fault finding. Explained fault clearly.	

9 - 10	Detailed system knowledge. Us documentation efficiently to find fault. Gave deta iled explanation of fault	
Timing (10%)		
Score	**Justification**	**Remarks**
10	Found Fault in allotted time	
8	Found Fault with an extra 15 minutes	
6	Found Fault with an extra 20 Minutes	
4	Found Fault with an extra 25 Minutes	
2	Found Fault with an extra 30 Minutes	
Find the Fault (10%)		
Score	**Justification**	**Remarks**
10	Fault Corrected	

Score	**Weighting**	**Total**
Initialization	**X 1**	**%**
Test Equipment	**X 1**	**%**
Fault Finding	**X 3**	**%**
System Knowledge	**X 3**	**%**
Timing	**X 1**	**%**
Finding the Fault	**X 1**	**%**
	TOTAL	**%**

Assessor Signature:	**Trainee Signature:**

Enabling Check Instructions For Written Tests

PART I - ADMINISTRATIVE INSTRUCTIONS

Materials Required:

1. pens, pencils, stationary;

2. specified references;

3. test questions; 4. test answer key; and

5. suitable Location.

Arrangement and Procedures:

6. The Trainee shall complete a written test, open/closed book disclosed minimum 24 hrs prior to test;

7. Review the questions with Trainee (s) prior to start of test;

8. All ECs will be collected after time allocated in EO; and

9. Instructors shall ensure that all ECs are marked and returned by the next training day.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC is an (indicate type - open, closed book) test. It requires that you provide written answers, and for open book, specific references to questions;

2. The time allotted for this EC is allocated in EO;

3. The passing mark for this EC is 60%;

4. Part marks will be awarded for incomplete or partial answers;

5. IAW CSE SOPs, any Trainee will be disqualified, their exam discontinued and circumstances reported to the CSE Div Cdr if:

   a. are in possession of an unauthorized book or manuscript;

   b. are found to be copying paper of another candidate;

   c. permit their papers to be copied; or

   d. attempt to receive assistance. of any description.

6. Do you have any questions?

New format

PERFORMANCE/ENABLING CHECK

PC/EC Time: ### mins (concurrent/per trainee) + ## min debrief **ASSESSMENT ACTIVITY**: (Brief description of the assessment activity that students will do IOT prove they have met the Standard of the PO/EO). **Pass/fail criteria**: (identify success criteria) **ASSESSMENT RESOURCES:** • (Bulleted list of everything given to trainees for PC/EC) **IMPACT OF FAILURE:** Remedial Training / Re-test

INVIGILATOR NOTES:

(Course Name) – PC/EC XXX.XX Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. PASS / FAIL CRITERIA: Identify success criteria.

2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	**PASS**	**FAIL**	**COMMENTS**
1. Criteria			
(Success criteria taken from PO/EO standard)			
2.			
ADDITIONAL COMMENTS			
STRENGTHS:	**AREAS FOR DEVELOPMENT:**		
THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			

OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:
Trainee	Name:	Date:	Signature:

ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)

(Certificate of Qualification)

Version	Revision Date	Remarks
OJPR Version 1.0	DD / MM/ YYYY	New OJPR

Name and Initials	Rank	Service Number

Date Training Started	DD MMM YYYY
Date of Completion	DD MMM YYYY

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance or Enabling Check to the required standard, IAW the associated checklist in ANNEX I, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

PROGRESS REVIEW

5. Progress review shall be carried out quarterly by the Divisional Officer and Section Head (NCMs) or Training Officer (Officers).

Example OJPR record:

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI and Unit
PC/EC XXX.XX				DD/MM/YYYY	

OJPR Reviewed By OJT Supervisor (Rank, Position, Name)	Signature	Date
Section Head (NCMs) or HOD (Officers)		

TRAINING COMPLETION CERTIFICATE

Date training completed:	
Div Officer (NCMs) or CO (Officers)	Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed the OJT to the standards stated in the Name of QSP.

TE Cmdt/CO	Date

ANNEX K – TT PLAN

PO/EO	PO/EO TITLE	EO	EC	PC	PO
PO 001	Maintain the Multi Ammunition Softkill System	72	1	18	91
EO 001.01	Describe the MASS	10			
EO 001.02	Initialize the MASS	7			
EO 001.03	Operate the MASS	51	1	18*	
EO 001.04	Perform Pre/Post Firing Check on MASS	4			
PO 002	Maintain Ship's Main Gun System	198	8	17	223
EO 002.01	Describe Bofors 57mm Gun System	5	1		
EO 002.02	Perform Pre/Post Firing Check on Bofors 57mm Gun System	14			
EO 002.03	Operation Bofors 57mm Gun Laying Unit (GLU) and Gun Control Panel (GCP)	15	1		
EO 002.04	Perform Close up Clear away and Misfire procedures on the Bofors 57mm	48	1		
EO 002.05	Maintain Bofors 57mm Power Distribution System	20	1		
EO 002.06	Maintain Bofors 57mm Hydraulic System	10	1		
EO 002.07	Maintain the Bofors 57mm Loading and Firing System	35	1		
EO 002.08	Maintain the Bofors 57mm Laying System	31	1		
PO 002.09	Maintain Ship's Main Gun System	20	1	17*	

PO 003	Maintain the Close In Weapon System	316	1	34	351

PO/EO	PO/EO TITLE	EO	EC	PC	PO
EO 003.01	Describe Close In Weapon System	6			
EO 003.02	Initialize Close In Weapon System	8			
EO 003.03	Maintain Close In Weapon System Peculiar Support Equipment	7			
EO 003.04	Maintain the Close In Weapon System Power Supply and Control Group Subsystem	22			
EO 003.05	Maintain the Close In Weapon System Environmental Control Group	22			
EO 003.06	Maintain the Close In Weapon System Built In Test Equipment	18			
EO 003.07	Maintain the Close In Weapon System Weapon Control Group	17			
EO 003.08	Maintain the Close In Weapon System Operator Control and Display Subsystem	12			
EO 003.09	Maintain the Close In Weapon System Gun Subassembly	64			
EO 003.10	Maintain the Close In Weapon System Mount Servo Subsystem	28		17*	

EO 003.11	Maintain the Close In Weapon System Radar Subsystem	54			
EO 003.12	Maintain the Close In Weapon System Surface Mode Subsystem	44	1	17*	
PO/EO	**PO/EO TITLE**	**EO**	**EC**	**PC**	**PO**
EO 003.13	Perform Pre/Post Firing Check on Close In Weapon System	14			
PO 004	Operate the CIWS	20	8		28
EO 004.01	Operate the CIWS	20	8*		
PO 005	Maintain Combat Management System 330	30	5		35
EO 005.01	Describe the CMS	10			
EO 005.02	Initialize the CMS	10			
EO 005.03	Repair the CMS	10	5		
PO 006	Utilize DRMIS Portal as a Maintenance Supervisor	4.4		0.7	5.1
EO 006.01	EO 006.02 and EO 006.03 Utilize DRMIS Portal as a Maintenance Supervisor	4.4		0.7	

ANNEX L – RECORD OF DECISIONS TPWB

1. This record summarizes the conduct of the TPWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title				
Location of Board				
Dates				
Board Composition				
	Surname	Initials	Rank	Unit
Co-Chair (content)				
Co-Chair (process)				
NFS				
SME				
SME				
SME				
SME				
SME/Secretary				
Advisor				

Day #	Details	Action
	The following information on target population was compiled during the initial steps of the TPWB to give context for design and development: (rank, experience, entry level education, etc.).	
1		
2		

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Amendment Form, thus ensuring the QSP has the most relevant information.

Version:	**X.X**
Date:	**Who Initiated Change:**
Requirement for the Amendment:	
Description of Changes:	
Change Tracking number:	

Trg Mgr:

ANNEX M – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

ACRONYM / ABBREVIATION	TERM/DEFINITION
AAW	Anti-Air Warfare
AC	Alternating Current
BIT	Built In Test
BITE	Built In Test Equipment
CMS	Combat Management System
CFTO	Canadian Forces Technical Order
CM	Corrective Maintenance
CSAT	Combat System Alignment Test
CSIT	Combat System Integration Test
CSOT	Combat System Operability Test
DC	Direct Current
DWAN	Defence Wide Area Network
ECG	Environmental Control Group
ECM	Electronic Counter Measures
ELX	Electronic Enclosure
EOSS	Electro-Optical Surveillance System
FCS	Fire Control System
FIT	Fault Isolation Test

HMI	Human Machine Interface
HOLC	High Order Language Computer
HV	High Voltage
IR	Infrared Radiation

ACRONYM / ABBREVIATION	TERM/DEFINITION
JBS	Job-Based Specifications
LAN	Local Area Network
LCP	Local Control Panel
LCS	Local Control Station
LRU	Lowest Replaceable Unit
MPT	Maintenance Procedural Trainer – Can be any form of computer based training, table top faults, or any other form of fault finding practice necessary.
MRC	Maintenance Requirement Card
MSDD	Mass Storage Disk Drive
PM	Preventative Maintenance
POFA	Programmed Operational Functional Analysis
PPE	Personal Protective Equipment
PSE	Peculiar Support Equipment
PSOT	Periodic System Operability Test
PsuM	Phalanx Surface Mode
RADZONE	Radiation Zone
RCP	Remote Control Panel
RCS	Remote Control Station

RF	Radio Frequency
RSMU	Radar Servo Maintenance Unit
RSI	Radar Servo Interface
SAGA	Stand Alone Gun Assembly
SHIPLAN	Ship Local Area Network
SOT	System Operability Test

ACRONYM / ABBREVIATION	TERM/DEFINITION
TCU	Transmitter Control Unit
TDP	Transmitter Data Processor
TPSU	Test Point Selector Unit
UPS	Uninterruptable Power Supply
WCC	Weapon Control Computer
WIC	Weapons Interface Cabinet